

PROVISIONAL PATENT APPLICATION

Force-Free-Radius and Measured-Weight Control Apparatus Using a Resonantly Driven Ordered Medium, Calibrated Modules and Plural-Module Systems, and Methods of Operating, Qualifying, and Recalibrating Them

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This specification is complete and self-contained. Applicant's companion applications are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no experiment, nonzero response, or qualified unit is represented as achieved, and no response, sign, magnitude, additivity, retained state, or scaling law is presumed.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media (companion drafts: transportation and lift systems; measurement instruments; energy storage and infrastructure; media and array architectures; information and winding-register devices; time-rate metrology; and completion-scheduled computation). Each companion states its own relationship to this disclosure and must remain self-contained as to claim-essential matter when filed.

FIELD

Support-force (weight) control apparatus and systems built on a resonantly driven ordered medium, and methods of operating them. More particularly: (i) a **support-force control apparatus** comprising a body of ordered medium, an alignment-field source, a resonant coupling structure and locked drive, a calibration memory holding an empirically established and independently validated support-force response map, and a setpoint controller that commands a selected change in the body's support force—and thereby in the measured weight of a load rigidly borne with it—from that map, within stored runtime limits and with safe-off and recalibration behavior; (ii) sealed **calibrated support-force modules** and **plural-module support-force and lift systems** operating from per-module maps and a measured array allocation; (iii) the **deployed operating method** by which an ordinary operator commands such apparatus from its stored map; and (iv) the **factory qualification, calibration, and commissioning** apparatus and methods—a null-valid resonance-drive and force-metrology protocol with preregistered acceptance gates—that establish the response map and operating envelope for a unit or production class.

Within the applicant's physical hypothesis, a commanded change in support force is equivalently a change in the radius between the centers of extent of two or more masses. Whether any unit or class becomes a qualified responsive module is decided by measurement, not presumed; the qualification apparatus remains operative and useful whether the measured response is nonzero or null.

BACKGROUND

1. No practical technology exists for adjusting the supported weight of an object on command. All existing lift and support technologies exchange momentum with a medium (aerodynamic, hydrodynamic, propulsive) or apply counter-forces (magnetic levitation against a prepared track or diamagnetic susceptibility; acoustic and optical trapping at micro scale). None modify the constraint force on the body itself, and none provide a selectable sign (heavier/lighter) of a genuine weight change on an ordinary force sensor in an unprepared environment.
2. The prior record contains repeated, partial approaches to spin-associated force effects:
 - Einstein–de Haas (1915) and Barnett (1915) established, to metrological standard, that alignment of internal circulation (extant label: electron spin/magnetization) and bulk mechanical rotation are one conserved ledger, exchangeable in both directions.
 - Wallace, U.S. Pat. Nos. 3,626,605 and 3,626,606 (1971), claimed force-field generation ("kinemassic field") from mechanically rotated bodies of half-integer-spin material. The approach relied on bulk mechanical rotation (alignment fractions of order 10^{-10} by the Barnett effect at practical speeds) with no resonant drive, no coherence architecture, no sign-resolving differential protocol; the claims were never reduced to accepted practice.
 - Laithwaite (Royal Institution, 1974) reported gyroscopic weight anomalies from bulk rotation; careful weighing did not sustain a static weight-loss claim. Bulk rotation is the wrong lever for the same reason as Wallace: negligible aligned fraction and rate.
 - Alzofon (AIAA-81-1608, 1981) proposed and partially tested support-force modification by pulsed dynamic nuclear orientation at GHz in Al/Fe composites — the closest art. The nuclear channel offers long relaxation but requires impractical polarization at room temperature; the program lacked a coherence architecture, a lock protocol, and a sign-resolved validity discipline. Later replication attempts at industrial laboratories reported anomalous motion of a test mass without a controlled differential protocol.
 - U.S. Pat. No. 10,144,532 B2 (2018) claims a craft using an "inertial mass reduction device" via high-frequency electromagnetic vibration; it teaches neither an aligned angular-momentum population, nor a resonance-locked coherent phase advance, nor a sign-selection mechanism, nor a differential force protocol.
3. What the art lacks — and this application supplies — is a support-force control device architecture in which (i) an already-aligned internal angular-momentum population in a medium of measured phase coherence is (ii) driven by a pulsed or continuous, coupling-structure- enhanced, frequency-locked drive at a measured precession resonance, and (iii) the device's support-force response to its

commanded drive state is not presumed from theory but is established, per unit or production class, by a factory qualification protocol having an independently supported force channel, randomized state sequencing, magnetic-gradient mapping, absorbed-energy-matched controls, reaction-force monitoring, sample and reference swaps, and preregistered acceptance gates; the resulting validated response map is stored in the device, and a setpoint controller operates the device from it. The architecture is designed to be built, qualified, and operated without presuming that the response is nonzero, gravitational, or of either sign: a unit whose map contains only null operating points is a qualified null reference; a unit whose map contains at least one independently validated nonzero operating point is a qualified responsive module operable at setpoints within its map. Constitutive mathematics relating drive state to support force is expected to follow the qualification measurements, as is ordinary for an empirically calibrated actuator; the absence of a closed-form transfer law before qualification does not narrow the device disclosed.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background teaching only: no material essential to enablement, written description, or the claims is incorporated from them, and this specification is complete and self-contained. The corpus may serve the examiner as an extended tutorial on the framework's empirical anchors and vocabulary.

SUMMARY

The disclosure is organized in three tiers (FIG. 3). The product and its deployed operating method lead; the qualification laboratory establishes the map the product runs from; the validation, recalibration, and fail-safe provisions govern what a deployed unit may command.

Tier 1 — the product and its deployed operating method (leading). A support-force control apparatus (400; FIG. 1) comprises: a body of ordered medium (402); a support interface (410) through which the body's support force is transmitted to a supported or supporting structure—a payload, platform, vehicle structure, load-bearing frame, or force sensor; an alignment-field source (404) establishing a measured alignment state of the body's internal angular-momentum population; a coupling structure (406) and locked drive chain (408) applying an oscillating drive at the body's located precession resonance; a reaction frame (422) carrying the alignment-field source and coupling structure on a load path separate from the support interface; a calibration memory (412) holding a **validated support-force response map** for the unit or its production class—an empirically established relation between drive state and support-force change containing at least one independently validated nonzero operating point, its uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters; runtime witness sensors (416); and a **support-force setpoint controller** (414) that receives a support-force setpoint, selects and applies the mapped drive state within the permissible interpolation region, regulates by feedback from a support-path force sensor (420) and/or feedforward from the map, monitors the runtime

witnesses against the stored limits, reports the achieved change with propagated uncertainty, and transitions to a safe-off state (418) and flags recalibration when a limit or validity condition fails. In the **deployed operating method** (FIG. 2) an ordinary operator commands the apparatus from its stored map; the operator does not perform the qualification laboratory.

Tier 2 — factory qualification, calibration, and commissioning. The response map is established, once per unit or production class, by a null-valid **qualification apparatus and protocol** (E1; FIGS. 7–10 and 12): the body on a sample force sensor mechanically separate from the coupling structure, alignment-field source, feed lines, and drive hardware; randomized and optionally blinded drive, control, reference-material, position-swap, and absorbed-energy-matched sham states; synchronized logging of support force, reaction force, drive power, temperature, pressure, vibration, acceleration, strain, magnetic field, and field gradient; and preregistered acceptance gates A1–A9 including independent replication. The output is an empirical response map with uncertainty and an artifact classification—including a valid null. A **qualified responsive module or product class** is one whose stored map contains at least one independently validated nonzero operating point; a unit whose map contains only null points is qualified as a **null reference** and accepts test-state commands only. Array qualification of an assembled plurality of modules produces a measured array allocation map in the same way.

Tier 3 — validation, interpretation, recalibration, and fail-safe. Validity states (valid, degraded, out-of-envelope, invalid, safe-off); runtime zero and reference checks; drift-triggered envelope shrinkage or recalibration; a service path back to full qualification; and the rule that a deployed unit never commands an operating point its map does not support.

Layered scope (FIG. 4). Every quantity is disclosed at three layers: (L0) the raw synchronous observable—the support-force change synchronized with a commanded drive state, always reportable; (L1) the validated response map—the map-corrected support-force change with uncertainty, reportable inside the validity envelope; and (L2) controlled operation—setpoint control of support force from the map. No layer presumes sign, magnitude, additivity, retained winding, or scaling; the map admits zero.

Products. (a) The support-force control apparatus and its operating method; (b) a sealed **calibrated support-force module** (300; FIG. 5) with a standard mechanical, electrical, and data interface; (c) a **plural-module support-force system** (600; FIG. 6) in which modules holding individual maps bear a common load under a system controller whose allocation across modules is taken from a measured array allocation map rather than from presumed additivity; and (d) prophetic operating species—payload trim, time-varying support-force profiles, station-keeping, soft landing, and measured array allocation—each conditional on a qualified module class whose map covers the required operating point.

The reference hardware is a single-crystal yttrium-iron-garnet body at a mapped transverse microwave magnetic-field antinode of a rectangular TE₁₀₂ coupling structure, an independently supported C-frame electromagnet, a locked pulsed drive chain with an enforced duty bound, independent sample and reaction-force sensors, environmental sensors, and a controller. No nonzero magnitude, sign, scale law, retained state, or transfer law is represented as experimentally established in this application; the transfer relation R is declared open (T6) and is completed empirically, per unit or class, at qualification.

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Force-free-radius and measured-weight control apparatus (400): body of ordered medium on a support interface to a supported structure; alignment-field source; coupling structure and locked drive chain; reaction frame on a separate load path; calibration memory holding the validated response map; support-force setpoint controller with feedback and feedforward paths; runtime witness sensors; safe-off and recalibration interface.
- FIG. 2 — Deployed operating method and runtime validity state machine: load map and verify provenance → zero check → receive setpoint → check permissible interpolation region → apply mapped drive state → feedback/feedforward regulation → monitor runtime witnesses → report achieved change → safe-off or recalibrate; states valid, degraded, out-of-envelope, invalid, and safe-off.
- FIG. 3 — Qualification-to-deployment flow in three tiers: factory qualification protocol with gates A1–A9 producing the empirical response map; classification into qualified responsive module or null reference; storage of map, provenance, interpolation region, runtime limits, and safe-off parameters; shipment; deployed operation; runtime checks; recalibration loop.
- FIG. 4 — Layered scope: L0 raw synchronous observable → L1 validated response map → L2 controlled operation, with the acceptance gates between L0 and L1 and the runtime validity gate between L1 and L2.
- FIG. 5 — Calibrated support-force module (300): sealed module with medium, coupling structure, alignment source, locked drive, embedded controller, calibration memory, witness sensing, and standard interface; setpoint mode enabled only for a qualified responsive module; null-reference mode otherwise.
- FIG. 6 — Plural-module support-force system (600): N modules with individual maps bearing a common load-bearing structure; common phase reference with per-module trim; system controller with measured array allocation map; system-level force and reaction accounting; array qualification states.
- FIG. 7 — Qualification apparatus block diagram: driven body on an independent sample force sensor within a separately supported resonator and static-field source; reaction-force sensor; VCO → pulse modulator → power amplifier → directional coupler → resonator; detector → controller; controller → field-polarity control and data log.
- FIG. 8 — Coupling detail: mapped transverse microwave magnetic-field antinode of a TE-mode, loop-gap, re-entrant, waveguide, coil, or mapped off-axis resonator; dielectric support stem mechanically independent of the coupling structure.
- FIG. 9 — Timing and state diagram: nonoverlapping pulse train; randomized ON/OFF, resonant/detuned, and field-sense windows; locked analysis plan.
- FIG. 10 — Qualification controller state machine and control matrix with acceptance gates A1–A9.
- FIG. 11 — Empirical response map as stored: the field-sense and drive-polarization response cells with uncertainty, learned after validation rather than assumed from a channel label; validated

operating points and the permissible interpolation region.

- FIG. 12 — Multi-timescale diagnostic separating cavity decay, population coherence, force response, thermal response, and mechanical response after drive termination.
- FIG. 13 — Nested rotating-frame qualification apparatus: Earth-reference baseline, displaced turntable rotation, and centered constraint-frame rotation; synchronized angular position, angular rate, angular acceleration, orientation, support-force, reaction-torque, state-amplitude, and response-latency channels; rotating-frame energy and candidate radial constitutive relation labeled at their respective evidentiary strengths.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Throughout this specification, extant-framework labels appearing in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names; their use commits neither the specification nor the claims to any entity ontology historically associated with those names. All claim terms take the operational definitions below; every measured claim quantity is fixed by a recited instrument reading. Hypothesis parameters are identified expressly and are not presumed to have independently measured values.

1. **Internal circulation / internal angular-momentum population** — the ensemble of constituent angular momenta of a medium; the quantity the extant literature labels "electron spin" and measures in aggregate as magnetization (A/m; standard magnetometry).
2. **Alignment field** — the static bias field (oersted or tesla) applied to establish or vary a measured alignment state of said population; preferred reference operation uses measured saturation alignment, and "alignment sense" is the field polarity.
3. **Precession resonance** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance, FMR), located operationally as the absorption extremum to which the drive servo locks.
4. **Coherent phase advance** — drive-synchronous periodic advance of the collective precession phase of an aligned population while phase coherence persists. It is specified by the measured drive frequency, phase lag, precession amplitude or cone angle, and the number of cycles within a continuous coherence interval. Cycle count alone is not presumed to be a retained material state after the drive terminates.
5. **Drive-state vector (x)** — the independently measured operating variables for one observation, including at least field magnitude and sense, drive frequency and phase, incident and absorbed

power, pulse width, repetition rate and duty factor, resonance mode, magnetization or aligned fraction, sample temperature, pressure, orientation, time, and, when an apparatus-rotation control is performed, the angular rate, angular acceleration, and sense of the local constraint frame relative to the body.

6. **Empirical response characteristic (R)** — the preregistered estimate of differential support force as a function of the drive-state vector and control state, together with its uncertainty, validity range, hysteresis, time constants, and null-control residuals. A characteristic is not treated as a weight-control calibration unless it has passed the commissioning gates and an independent replication.
- 6A. **Channel partition (λ)** — a hypothesis parameter describing how driven angular momentum may be divided between intrinsic-alignment ("twist") and orbital-winding ("writhe") bookkeeping. It is not defined by the support-force result, is not required to operate the reference experiment, and is not used as proof that a measured response has a particular physical cause.
7. **Support force / measured weight** — the force read by a sensor supporting the body (newton), under the differential protocol of this disclosure. 7A. **Force-free radius** — the radius or local curvilinear path on which a body follows its unforced motion while a co-moving sensor registers zero support force. A measured-weight change at a fixed support interface is the changed constraint reaction associated, under the applicant's hypothesis, with holding the body away from its force-free radius. The term does not presume a force transmitted across otherwise empty space and does not assert arbitrary control of optical distance between unconstrained bodies.
8. **Rate state / proper-time rate** — the body's internal rate as read by a co-located clock; "differential rate" is the fractional frequency difference $\Delta f/f$ between co-located and reference clocks (companion time-rate application).
9. **Quanta rate ($\dot{N}$)** — delivered power expressed as a count rate of action quanta, $\dot{N} = P/h\nu$, at drive turn rate ν .
10. **Turn rate (ν)** — frequency in cycles per second (hertz); energy and rate related by $E = h\nu$.
11. **Linkage / sink readiness** — the condition in which a resonant load ("sink") is presented and matched such that delivery can proceed (extant labels: matched load, critical coupling); "sink-gated" operation enables delivery only upon such presentation.
12. **Power on the wire / delivered power** — wherever this specification speaks of power delivered, transported, or present on a wire, feed, or port, the referent is the measured power (watts) at the stated port; no carrier-substance picture is asserted or required.
13. **Oscillating drive** — a time-periodic influence applied to the body at a stated turn rate through a coupling structure (cavity, coil, or antenna), specified operationally by its frequency, delivered power, and pulse timing at the stated port (extant labels: radio-frequency, microwave, or electromagnetic drive).
14. **Population coherence time** — the decay time of the coherent material precession, determined independently from the material resonance linewidth or a time-domain pulsed magnetic response (extant label: transverse relaxation time T_2^*). It is distinguished from the cavity energy-decay time and from any slower mechanical, thermal, or force-sensor response.

15. **Sufficient phase coherence** — the condition that the population coherence time τ and the drive turn rate ν satisfy $\tau \cdot \nu \geq 1$: the driven population retains coherence for at least one full turn of drive phase. A "medium (or material) of sufficient phase coherence" is one satisfying this condition under the recited drive and alignment field. Preferred embodiments exceed this floor by two to three orders of magnitude (E4; single-crystal yttrium iron garnet at gigahertz-class drive may provide $\tau \cdot \nu$ of order 10^2 – 10^3 , to be measured for the selected specimen and mode).
16. **Cavity energy-decay time** — the independently measured decay time of energy stored in the coupling structure, determined from its loaded quality factor and ring-down response; it is not identified with population coherence time.
17. **Absorbed-energy-matched control** — a control state adjusted so the active and control runs have matched absorbed-energy and temperature histories within a stated tolerance, using incident, reflected, and transmitted power and, where practicable, calorimetry; equality of source power alone is not sufficient.
18. **Validated support-force response** — a nonzero empirical response characteristic that passes all preregistered artifact-rejection gates, remains after correction for the declared search space and multiple comparisons, repeats after sample/reference and position swaps, and is independently reproduced on a second run, apparatus, or laboratory. A synchronous balance deflection that has not passed these gates is an anomaly, not a demonstrated change in measured weight.
19. **Empirical response map (stored)** — the data structure held in a calibration memory for a unit or production class: the estimated empirical response characteristic R of Definition 6 evaluated over a set of drive states, giving at each state the change in support force relative to the drive-off baseline, its uncertainty or covariance, hysteresis and time constants, thermal and orientation coefficients, and the identity of every operating point that was independently validated under Definition 18; together with the provenance record, validity envelope, runtime limits, and safe-off parameters of Definitions 20–22. A map may contain only null operating points. The map is established by measurement of the unit or class; it presumes no sign, magnitude, additivity, retained state, or scaling beyond its validated points.
20. **Provenance record** — the identity of the unit or production class; the qualification apparatus, reference bodies, and force references used; date; operator or process; raw-data identifier; fit method; replication record; and the acceptance gates passed for each validated operating point.
21. **Validity envelope; permissible interpolation region; runtime limits** — the region of drive-state and environmental variables over which the map was established and validated, including the region within which interpolation between validated operating points is permitted; together with the runtime limits (temperature, orientation, force-axis field gradient, vibration and acceleration, absorbed power and duty, bandwidth and slew, drift rate, zero-check interval) outside which the controller reports out-of-envelope. Extrapolation beyond the region is not permitted.
22. **Safe-off** — the stored state to which the controller drives the apparatus on a validity failure, fault, or command: oscillating drive off; alignment field to a stored safe magnitude and sense (or unchanged in a fixed-bias unit); last valid record preserved; recalibration flag set.

23. **Qualified responsive module; qualified product class; null reference** — a unit, or a production class represented by qualified units, whose calibration memory holds an empirical response map containing at least one independently validated nonzero operating point under Definition 18, its uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters, is a *qualified responsive module* (or *class*). A unit whose map contains only null operating points is a *null reference*: it accepts test-state commands and reports L0 and L1 quantities but has no enabled setpoint mode. Neither status presumes any response, sign, magnitude, additivity, or scaling beyond the validated points.
24. **Support interface** — the mechanical path through which the body's support force is transmitted to a supported or supporting structure: a payload, platform, vehicle structure, load-bearing frame, or force sensor. The body's support force and the reaction loads of the alignment-field source and coupling structure pass through separate load paths.
25. **Support-force setpoint** — a commanded change in the body's support force relative to its drive-off baseline, expressed in newtons with sign, as a fraction of the baseline support force of the body or of a load rigidly borne with it, or as a time-varying profile of such values within a stored bandwidth and slew limit.
26. **Runtime witnesses** — the reduced set of sensor readings monitored during deployed operation and compared with the runtime limits: at least body and mount temperature, orientation, a force-axis field-gradient witness, acceleration and vibration, incident, reflected, and absorbed drive power and duty, lock state, and, where present, support-path force and reaction-frame force.
27. **Validity states** — the controller-reported states *valid* (setpoint mode permitted), *degraded* (L0 and L1 reported; setpoint held or reduced; update required), *out-of-envelope* (a witness lies beyond a runtime limit; setpoint inhibited), *invalid* (map check or gate failure; recalibration flagged), and *safe-off* (Definition 22), as shown in FIG. 2.
28. **Array allocation map** — for a plural-module system, the measured relation between a set of per-module drive states and the change in the system-level support force of the assembled plurality, established under sequential, in-phase, phase-scrambled, and inactive-module qualification states, with uncertainty and cross-talk terms. The sum of individual module responses is not presumed.

THEORETICAL FOUNDATION AND OPEN TRANSFER QUESTION

The applicant provides the motivating framework so that the physical hypothesis, the variables to be measured, and the experiment capable of rejecting the hypothesis are stated in one self-contained disclosure. The framework is the Pope–Osborne Angular Momentum Synthesis (POAMS; N. V. Pope and A. D. Osborne, 1970s–2000s, published record), with the applicant's extensions. The conventional operation of the resonance, sensing, control, and data-acquisition apparatus does not depend on acceptance of that framework. No nonzero support-force effect, sign, magnitude, material scaling law, or

phase-to-force transfer law is represented as measured or established. The open transfer question is completed empirically, per unit or production class, by the qualification stage disclosed below; the product operates from the resulting stored map, not from a derived coefficient.

T1. Units and rates. The constant c is a units conversion between observed distance and observed time (empirical anchor: the operational definition of the metre). Energy is turn rate: $E = h\nu$, with h the action of one completed turn of internal phase (h per radian). Mass is trapped rate: $m = E/c^2 = h\nu/c^2$. One atomic mass unit corresponds to $\bar{\nu} = uc^2/h = 225.2$ ZHz of internal turning; a 56-quantum iron atom carries ≈ 12.6 YHz aggregate. What a balance registers as mass is, quantitatively, h/c^2 times the body's total internal turn rate. These identities are standard algebra on measured constants; they commit the examiner to nothing exotic. Two features of the SI itself anchor them in standards practice. First, time: since 1967 the SI second is a count of internal turning — 9,192,631,770 cycles of the cesium-133 hyperfine transition — and that transition is itself a rate difference between two organizations of the atom's angular momentum ($\Delta E = h\nu$ read in reverse); the same atom's full internal rate is $m_{\text{Cs}} c^2/h = 2.99 \times 10^{25}$ turns per second, so the world time standard already reads time as counted turning. Second, energy-per-charge: since the 2019 SI redefinition the volt is realized through the Josephson constant $K_J = 2e/h = 483,597.848$ GHz per volt — national laboratories hold the volt by counting a turn rate against a junction array; equivalently, one elementary charge moved through one volt corresponds to $e/h = 241.799$ THz of turn rate (bench receipt: a red light-emitting diode's ≈ 1.9 V forward drop times e/h is ≈ 460 THz — the frequency of red light; undergraduate laboratories measure h this way). $E = h\nu$ is therefore not the applicant's postulate: it is the arithmetic backbone of modern metrology. The applicant's further interpretation of mass and time as organized turning is a motivating ontology; the SI identities alone do not establish that a resonant drive can alter support force.

T2. Weight as constraint force (hypothesis under test). In the POAMS reading of orbital mechanics (compared against the measured record: Mercury perihelion advance; GPS clock offsets; the factor-three orbital time-dilation relation of Osborne & Pope), a body free at radius r with angular momentum L follows its natural path; supporting it elsewhere requires a constraint force — the support force a scale reads (as differentiated from any purely theoretical invisible in-vacuo force of attraction or repulsion). Weight is thus a bookkeeping entry on the body's angular-momentum state, not an immutable property: a body whose internal angular-momentum state is altered has, quantitatively, an altered natural path and an altered constraint bill.

A geometric corollary grounds the language of the Field statement above. Within this framework every distance between masses is a radius: the straight line is the limiting case — a single theoretical radius of unlimited extent — and rectilinear force-free motion, assumed since Newton, has nowhere been observed; force-free natural motion is curvilinear. A force-free body is therefore continuously accelerating in the kinematic sense (its velocity direction turns) while a co-moving sensor reads zero force — and the extant record contains this fact wholesale, since every orbiting body accelerates continuously toward its primary while its instruments read weightlessness. Acceleration without felt force

is precisely the distinction between path geometry and constraint force on which this specification rests, and amending "the radius between the centers of extent" of two masses (Field statement) is accordingly the literal object of the method, not a figure of speech.

The coupling "constant" G is hypothesized in this framework to depend on the body's internal rate state. The GPS receipt deserves its numbers, because an examiner can verify them from public documents in minutes. The orbital-clock relation gives $d\tau/dt = 1 - (3/2)GM_{\oplus}/rc^2 = 1 - 2.5045 \times 10^{-10}$ at the GPS radius $r = 26,562$ km; the rotating-geoid reference runs at $1 - GM_{\oplus}/R_{\oplus}c^2 - v_{\text{spin}}^2/2c^2 = 1 - 6.973 \times 10^{-10}$; the resulting net is approximately $+4.46 \times 10^{-10}$, or approximately $+38.5$ $\mu\text{s/day}$. The constellation's published remedy is a manufactured oscillator-frequency offset: each nominal 10.23 MHz standard is built to run at 10.2299999954326 MHz on the ground. The two-line approximation agrees with the published correction in sign and scale; it is not used as evidence for the laboratory transfer law. The same ledger at $r = 6,791$ km returns -24.4 $\mu\text{s/day}$ for the International Space Station (≈ 4.4 ms of proper-time deficit per six-month crew rotation, matching flown-clock accounting), with the sign crossover at $r = (3/2)R_{\oplus} \approx 3,200$ km altitude. These established translational and orbital relations motivate the inquiry, but do not establish the proposed bridge from coherent internal precession in a stationary medium to support force or proper time. That bridge is the express experimental question of this disclosure.

T3. Alignment is mechanical angular momentum. Einstein–de Haas and Barnett measured the two directions of the same ledger: align a body's internal circulations and it rotates; rotate it and they align. The internal circulations aligned by a static field are mechanical angular momentum at the constituent level — order-unity alignable at saturation in the garnet-class media of this disclosure, versus $\sim 10^{-10}$ by bulk rotation at practical speeds (the quantitative reason Wallace- and Laithwaite-class approaches read null). A modern industrial receipt now supplements the 1915 metrology: spin-transfer-torque magnetic memory (STT-MRAM), shipping commercially, switches its storage layer by direct angular-momentum delivery, and the extant art's own write equation is conservation of angular momentum, $dL/dt = \eta(\hbar/2)$ (I/e) — each conduction carrier (the extant art's "electron") hands the layer up to $\hbar/2$. On public device values (a $40 \text{ nm} \times 1.3 \text{ nm}$ CoFeB free layer at $M_s = 10^6$ A/m holds $\approx 1.8 \times 10^5$ units of $\hbar/2$; a $50 \mu\text{A}$ write current delivers carriers at $I/e = 3.1 \times 10^{14} \text{ s}^{-1}$ at transfer efficiency $\eta \approx 0.6$) the conservation arithmetic returns $t_{\text{switch}} = N \cdot e / (\eta I) \approx 0.9 \text{ ns}$ — the datasheet class of the shipping devices. An industry writes billions of bits per second by exactly the accounting this section states: aligned spin populations are mechanical angular momentum, deliverable and conserved at device level.

T3A. Rotating-frame alignment and nested local centers. For a body carrying angular momentum $\mathbf{L}$ in a frame rotating at angular rate $\boldsymbol{\Omega}$, the standard rotating-frame energy contains the term

$$U_{\text{rot}} = -\boldsymbol{\Omega} \cdot \mathbf{L},$$

and the corresponding orienting torque is $\boldsymbol{\tau} = \boldsymbol{\Omega} \times \mathbf{L}$. Parallel and antiparallel states differ by $2|\boldsymbol{\Omega}||\mathbf{L}|$ when the vectors are collinear; dissipation therefore permits an antiparallel state to turn toward the lower-energy parallel state. This relation supplies a macroscopic, sign-resolving bridge between angular-momentum orientation and a directly observable mechanical response, but it does not by itself prove a stationary change in measured weight.

The applicant has repeatedly observed qualitatively, using high-speed rotors in three-axis gimbals, that a rotor aligned in sense with the operative rotating frame remains stable, whereas the opposed state rapidly flips; reversing an imposed turntable rotation reverses the preferred orientation, and rotating the gimbal cage about a centered rotor produces the same sense selection without requiring the rotor center to complete an orbit. The observation is reported as empirical motivation, not as a calibrated result relied upon for claim utility. It motivates treating the operative angular rate as a nested local quantity, $\Omega_{\text{eff}} = \Omega_{\text{Earth}} + \Omega_{\text{apparatus}} + \Omega_{\text{frame}}$, and adding controlled apparatus/frame rotation to the qualification matrix. A locally imposed angular rate may dominate Earth's angular rate immediately; accumulated translational distance about the new center is not required for the rotating-frame energy term to change.

T4. Coherent phase advance. With the population aligned, a resonant drive at the precession frequency (extant label: FMR; $f = \gamma H_{\text{eff}}$, $\gamma \approx 2.8 \text{ MHz/Oe}$ electron-class) maintains a collective phase relation and precession amplitude. The number of completed phase cycles within one measured coherence interval is $\nu\tau_{\text{coh}}$. In ordinary steady-state FMR, orientation is periodic modulo 2π and additional cycles do not by themselves create an indefinitely growing stored state; any retained or cumulative state is therefore a separate observable that must be demonstrated rather than assumed. The commissioning variables include aligned fraction, precession amplitude, phase lag, $\nu\tau_{\text{coh}}$, absorbed energy, and population. Yttrium iron garnet ($\text{Y}_3\text{Fe}_5\text{O}_{12}$, YIG) is the reference material because it is an insulating ferrimagnet with high saturation induction and narrow, specimen-dependent precession linewidth.

T5. Candidate channels and sign test. The applicant's model distinguishes an intrinsic-alignment ("twist") bookkeeping channel from an orbital-winding ("writhe") channel and hypothesizes that they would contribute opposite support-force signs for a fixed global orientation. The topological identity $L_k = T_w + W_r$ motivates that vocabulary but is not asserted, by itself, to supply a quantitative mapping from a YIG spin ensemble to a closed ribbon or to prove a force law. The channel partition λ is not known independently and is not used to predict or validate a result. Field sense, drive polarization, sample orientation, and cavity mode are independent experimental factors. The direction and magnitude of any response are learned from the empirical response characteristic. Only after validation may one field sense be associated with a desired signed offset within the measured range.

T6. Open transfer relation and empirical completion. The application does not assume a known magnitude law. It declares the missing relation explicitly:

$$\Delta F_{\text{candidate}} = R(\text{Bo, sense, } f, \phi, P_{\text{abs}}, B_1, Q_L, D, M/M_s, \tau_{\text{coh}}, V, T, p, \text{orientation, geometry, } \Omega_{\text{frame}}, d\Omega_{\text{frame}}/dt, t),$$

where every argument is measured, D is duty factor, and R may be identically zero. The commissioning protocol estimates R with uncertainty and compares it to an explicit nuisance model for magnetic, thermal, mechanical, radiometric, electrical, and statistical effects. The applicant's hypothesis is that aligned population and coherent drive state provide the principal candidate lever rather than bulk rotation

speed. Neither a linear population law nor additivity across bodies is presumed. If a validated nonzero R is obtained, its measured sign, gain, bandwidth, hysteresis, saturation, spatial reach, and energy cost—not an assumed theoretical coefficient—define the usable transfer characteristic.

One dimensionally consistent candidate factorization, offered to organize measurement and not as an established law, begins with a coupling energy

$$U_c(\mathbf{r}) = -\lambda \Omega_{\text{nat}}(\mathbf{r}) \cdot \mathbf{L}_{\text{posted}},$$

where $\Omega_{\text{nat}}(\mathbf{r})$ is the angular rate associated with the local force-free path at radius $\mathbf{r}$, $\mathbf{L}_{\text{posted}}$ is the portion of the driven collective angular-momentum state that is externally coupled rather than returned through ordinary relaxation or the drive assembly, and λ is the open channel-partition parameter of Definition 6A and may be zero. The candidate radial constraint response is the generalized-coordinate derivative

$$\Delta \mathbf{F}_r = -\partial U_c / \partial \mathbf{r} = \partial [\lambda \Omega_{\text{nat}}(\mathbf{r}) \cdot \mathbf{L}_{\text{posted}}] / \partial \mathbf{r}.$$

For fixed local orientation and locally constant λ this reduces to $\Delta \mathbf{F}_r =$

$\lambda (d\Omega_{\text{nat}}/dr) \Delta \mathbf{L}_{\text{posted, parallel}}$, which has units of force. This equation does not establish the form of $\Omega_{\text{nat}}(r)$, a nonzero λ , linearity, or a cumulative state. It identifies the quantities a later derivation or empirical map must supply.

The posted state is likewise left open. A bookkeeping decomposition is

$$\mathbf{L}_{\text{posted}}(t) = \mathbf{L}_{\text{align}} + \int_0^t \eta_w(\mathbf{x}, t') \mathbf{A}_{\text{turn}}(\mathbf{x}, t') [\dot{\phi}(t') / (2\pi)] dt' - \mathbf{L}_{\text{relax}}(t),$$

where $\mathbf{L}_{\text{align}}$ is the finite static-alignment contribution, $\mathbf{A}_{\text{turn}}$ is an effective angular-momentum increment per coherently completed collective turn, η_w is a posting fraction that may be zero, and $\mathbf{L}_{\text{relax}}$ includes return through population relaxation, phase slips, heating, the coupling structure, and other measured reaction paths. The qualification experiment distinguishes: alignment-only saturation; a drive-rate response that vanishes at drive removal; a response following accumulated phase; a retained response with a measured decay constant; and a null. No one of those outcomes is presumed.

T7. Energy and momentum accounting. The controller meters incident, reflected, and transmitted drive power, estimates absorbed energy, logs thermal storage and dissipation, and records both sample-support force and reaction force on the separately supported drive assembly. A candidate response is rejected as a local actuator artifact when an equal and opposite reaction or another measured nuisance channel accounts for it. If a later calibrated embodiment performs mechanical work, that work is included with drive input, stored energy, heat, and all measured reaction channels in a complete energy balance. No particular conversion efficiency or restoration time is presumed before measurement; no perpetual-motion operation is disclosed or claimed.

T8. Constitutive-census hypothesis and scope. The framework distinguishes the exchangeable orientation and precession state of an aligned population from the constitutive angular momentum carried by the medium. The apparatus changes the former; it is not asserted to remove or consume the latter. This distinction motivates reversible drive protocols and return-to-baseline tests, but restoration time and completeness remain measured outputs. Einstein–Cartan spin-fluid models contain inverse-sixth-power

spin-density terms in remote high-density cosmological settings (Trautman, *Nature Phys. Sci.* **242** (1973) 7; Popławski 2010–2014). That analogy neither establishes a laboratory support-force response nor supplies R. No claim depends on cosmology or on the analogy being correct.

THE FORCE-FREE-RADIUS AND MEASURED-WEIGHT CONTROL APPARATUS (400) — PRODUCT ARCHITECTURE (FIG. 1)

Body of ordered medium (402) and holder (403). The reference body is a polished single-crystal YIG sphere, diameter 2–15 mm (reference 10 mm; volume 0.5236 cm³; mass approximately 2.71 g at 5.17 g/cm³; nominal drive-off baseline support force approximately 26.6 mN at standard gravity). Alternative media are those of E4, each qualified on its own measurements. The body is carried in a low-loss, nonmagnetic holder (quartz, sapphire, PTFE, alumina) that forms the upper element of the support interface.

Support interface (410) and optional support-path force sensor (420). The holder transmits the body's support force to the supported or supporting structure: in a bench or metrology product, to a force sensor; in a load-bearing product, rigidly to a payload deck, platform, or vehicle frame, so that a change in the body's support force appears as a change in the support force of the rigid assembly by ordinary statics. Because mounting class, orientation, and reaction paths are recorded drive-state variables (Definition 5), the map used in a load-bearing configuration is established or confirmed with the module in its mounting class and orientation; an offset measured on the bench is not presumed to transfer unchanged to a different mounting. Where a support-path force sensor (420) is present, it supplies closed-loop feedback and runtime zero checks; where it is omitted, the unit operates feedforward from the map with witness monitoring and interleaved drive-off zero checks read on the runtime witnesses.

Alignment-field source (404). An independently supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly producing a static field of commanded or fixed magnitude and sense, reference 1500–2500 Oe for the reference body. The qualification reference source is the 76 mm-pole C-frame electromagnet of E1 (about 0.196 T at approximately 20 A and an 80 mm gap; a roughly 0.20-kW thermal load); deployed units preferably use a permanent-magnet yoke with a trim winding for sense selection or a fixed-bias yoke, sized to the field the map requires and subject to the same gradient budget. The sign of support-force change associated with either sense is taken from the map, never from the field polarity.

Coupling structure (406) and drive chain (408). A TE-mode rectangular cavity (reference TE₁₀₂ at 5.600 GHz, empty-cavity seed approximately 47.55 × 22.15 × 64.77 mm), loop-gap resonator, re-entrant resonator, waveguide, or coil places the body at a mapped transverse microwave magnetic-field antinode with the drive's magnetic axis orthogonal to the alignment field; the loaded field vector is mapped after installation of body and holder. The drive chain comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), a pulse and burst modulator with a controller-enforced duty bound, a power amplifier sized to the map (reference 1–5 W at qualification), a directional coupler, and incident, reflected, and transmitted power detectors. The controller servo-locks to the located precession resonance and records mode, linewidth, loaded quality factor, and ring-up and ring-down times. Pulses are

nonoverlapping; higher delivered energy is implemented as bursts separated by recovery intervals or as thermally matched continuous-wave operation; $Q \times$ source power is not treated as a field or power multiplier.

Reaction frame (422) and reaction-force witness (424). The alignment-field source, coupling structure, feeds, thermal hardware, and shields are carried by a frame mechanically independent of the support interface, and their support loads pass through a reaction load path to the supporting structure or base. Where the map requires it, a reaction-force sensor on this path is a runtime witness: a commanded change in support force that is accounted for within tolerance by an equal and opposite change on the reaction frame is an actuator artifact, and the controller reports invalid.

Calibration memory (412). Holds the empirical response map, provenance record, validity envelope and permissible interpolation region, runtime limits, safe-off parameters, unit or class identifier, validity-state history, zero-check history, and recalibration schedule. The controller refuses a map whose provenance does not match the unit or class identifier.

Support-force setpoint controller (414). Receives a support-force setpoint (Definition 25); verifies that the setpoint lies within the permissible interpolation region and inhibits it otherwise; determines the corresponding drive state by inverting or interpolating the map; commands the alignment-field source and drive chain to that state within the stored slew and bandwidth limits; regulates by feedforward from the map and, where a support-path force sensor is present, by feedback, with loop gains bounded by the map's bandwidth, hysteresis, and time constants; compares the runtime witnesses (416) with the runtime limits on every control cycle; propagates uncertainty and reports the achieved change with its uncertainty; maintains the validity state (Definition 27); and executes safe-off (418). No sign, magnitude, or additivity is computed from a theoretical coefficient; every commanded state is a validated or permissibly interpolated point of the map.

Runtime witness sensors (416). Definition 26. The runtime set is a reduced subset of the qualification instrumentation of E1.

Safe-off and recalibration interface (418, 426). Safe-off per Definition 22. A service interface permits a reduced re-qualification (drive-off and sense-reversed zero checks and at least one validated operating point checked against a separate force reference) or return of the unit to the full qualification protocol of E1; a unit that fails the reduced check has its setpoint mode disabled until re-qualified.

Magnetic-gradient budget at runtime. For the saturated reference body (moment approximately $0.074 \text{ A} \cdot \text{m}^2$), a force-axis gradient of approximately $1.32 \text{ } \mu\text{T/m}$ imitates 98 nN . The runtime gradient witness enforces, as a runtime limit, the same rule applied at qualification: the residual force-axis gradient uncertainty stays below a preregistered fraction (preferably one third, approximately $0.44 \text{ } \mu\text{T/m}$ for the stated example) of the gradient that would imitate the smallest reportable support-force change; otherwise the state is out-of-envelope.

Energy and momentum account. Per T7, the controller logs incident, reflected, transmitted, and absorbed power, thermal storage and dissipation, support-path and reaction-path forces, and any mechanical work performed by a commanded change; no conversion efficiency, restoration time, or lift power is presumed before measurement.

DEPLOYED OPERATING METHOD (FIG. 2)

The deployed operating method is performed by the apparatus of FIG. 1 or a module of FIG. 5 whose calibration memory already holds a validated map. Its steps are: (a) *Power-on and self-test*. Load the map; verify that the provenance record matches the unit or class identifier; execute a drive-off zero check and, where the source permits, a sense-reversed zero check; compare with the stored baseline; enter *valid* only if all pass, otherwise *invalid* or *safe-off*. (b) *Receive setpoint*. Accept a support-force setpoint from an operator, a supervisory controller, or a stored profile. (c) *Check region*. Confirm that the setpoint and the current environmental state lie within the permissible interpolation region and runtime limits; otherwise inhibit and report *out-of-envelope*. (d) *Determine mapped state*. Invert or interpolate the map to obtain the drive state (field magnitude and sense, frequency, phase, power, pulse width, repetition rate or burst pattern, duty) and its predicted change with uncertainty. (e) *Apply*. Ramp the alignment field (if commanded) in a settled block and apply the locked drive within the stored slew and bandwidth limits. (f) *Regulate*. Feedforward from the map; where a support-path force sensor is present, close the loop on the measured support force with the stored gain bounds. (g) *Monitor*. On each control cycle compare the runtime witnesses with the runtime limits and, where a reaction-force witness is present, test that the commanded change is not accounted for by the reaction path. (h) *Report*. Output the achieved change in support force with propagated uncertainty and the validity state. (i) *Zero checks*. At the stored interval interleave drive-off (and, where permitted, sense-reversed) zero checks; drift beyond tolerance transitions to *degraded*, shrinks the permissible region, or flags recalibration. (j) *Safe-off*. On a runtime-limit exceedance, a witness deviation from the map beyond tolerance, loss of lock, or fault, transition to *safe-off* and set the recalibration flag. (k) *Recalibrate*. Perform the reduced re-qualification of the service interface or return the unit to full qualification. The operator of a deployed unit does not perform the qualification protocol of E1; gates A1–A9 are executed once at qualification and again only on service recalibration.

LAYERED SCOPE (FIG. 4)

- **L0 — raw synchronous observable.** The change in support force synchronized with a commanded change of drive state, $\Delta F_s = \text{mean}(F_{\text{ON}}) - \text{mean}(F_{\text{OFF}})$ at alignment sense s , or the in-phase and quadrature components at a chop or toggle frequency, each with statistical uncertainty from the recorded cycles and drift model. L0 is always reportable; it may be zero.
- **L1 — validated response map.** The map-corrected support-force change with uncertainty at a recorded drive and environmental state, reportable when that state lies inside the validity envelope.

The acceptance gates A1–A9 stand between L0 and L1: no operating point is a validated point of the map until it has passed them.

- **L2 — controlled operation.** Setpoint control of support force from the map by the controller of FIG. 1, the module of FIG. 5, or the system of FIG. 6. The runtime validity gate (Definition 27) stands between L1 and L2: a deployed unit commands only validated or permissibly interpolated points and only in the *valid* state.

QUALIFICATION, CALIBRATION, AND DEPLOYMENT (FIG. 3)

Factory qualification versus deployed operation. Qualification is performed once per unit or production class with the full instrumentation and control matrix of E1; deployed operation uses the stored map, the runtime witnesses, and the reduced self-tests of the deployed operating method. A deployed unit never commands an operating point its map does not support.

Empirical-map procedure. (a) *Acquire*: execute the randomized sequence of drive and control states of E1 across the intended envelope, recording all synchronized channels. (b) *Fit*: estimate $R(T6)$ as a function of drive state with a preregistered model class (lookup with interpolation, polynomial, or physics-agnostic regression), with covariance, hysteresis, and time constants, alongside the nuisance model. (c) *Validate*: classify each candidate operating point as null, identified artifact, or unexplained drive-correlated residual; apply gates A1–A8 to any nonzero residual; obtain independent replication (A9); test the fitted map on held-out operating points. (d) *Store*: write the map, covariance, provenance, permissible interpolation region, runtime limits, and safe-off parameters to the calibration memory, and record the qualification status—qualified responsive module if at least one nonzero operating point passed A1–A9, null reference otherwise. (e) *Ship and operate*: the deployed operating method of FIG. 2. (f) *Update*: runtime zero and reference checks are logged; drift beyond tolerance shrinks the region or triggers recalibration.

Production-class qualification. A class map is established from a plurality of qualified units of common design. An additional unit is accepted under the class map only after an acceptance test at one or more validated operating points against a separate force reference agrees with the class map within a stored tolerance; a unit that does not agree is qualified individually or as a null reference.

Array qualification. For a plural-module system the assembled plurality is qualified under sequential, in-phase, phase-scrambled, and inactive-module states with per-module and system-level force and reaction channels (E3), producing the array allocation map of Definition 28.

What qualification does not presume. Sign, magnitude, linear scaling with population or power, additivity across modules, retained winding after drive termination, and any lift or null-crossing power are outputs of measurement; each remains absent from the map until measured and validated.

DETAILED DESCRIPTION OF EMBODIMENTS

E1 — Qualification apparatus and null-valid qualification protocol (reference embodiment; FIGS. 7–10 and 12). In this specification "commissioning," "characterization," and "qualification" name the same factory stage. The apparatus of E1 is the Tier-2 qualification apparatus; it is also the reference hardware from which the product apparatus of FIG. 1 and the module of FIG. 5 are built, the product retaining the medium, alignment source, coupling structure, drive chain, reaction frame, and a reduced witness set, and adding the calibration memory and setpoint controller.

A polished single-crystal YIG sphere, diameter 2–15 mm (reference: 10 mm; volume 0.5236 cm³; mass approximately 2.71 g at 5.17 g/cm³), is placed at a mapped transverse microwave magnetic-field antinode of a coupling structure selected to excite the intended ferromagnetic-resonance or magnetostatic mode. The reference executable structure is a rectangular TE₁₀₂ cavity having empty-cavity seed dimensions approximately 47.55 mm × 22.15 mm × 64.77 mm for 5.600 GHz. The sphere is centered at the magnetic antinode, with the microwave magnetic field along a first horizontal axis and the static field along an orthogonal horizontal axis. A vertical quartz or sapphire stem passes through an approximately 3 mm below-cutoff aperture to the sample force sensor. In the ideal empty- cavity field map, the microwave magnetic-field magnitude remains at least approximately 88.5% of its central value across a 10 mm sphere and the cross-polar component remains below approximately 5.8%; the loaded cavity is tuned and the actual field vector is mapped after installation of the sample and stem.

Enabled alternatives include another TE-mode cavity, loop-gap resonator, re-entrant resonator, waveguide or coil structure, and an off-axis location in another resonator after the local microwave magnetic-field vector has been mapped. A cylindrical TM₀₁₀ resonator may be used only at a mapped off-axis location having adequate transverse microwave magnetic field: its axial center has electric-field maximum but microwave magnetic field zero and is not the reference sample location. The coupling structure, alignment-field source, coupling probe, feed lines, thermal hardware, and shields are carried by a mechanically independent frame and do not rest on the sample force sensor. A second force sensor under that frame records its reaction channel. An optional outer platform sensor records the external support force of the complete test head; otherwise the calibrated sum of all instrumented support paths supplies that account. Feed lines are strain-relieved on the independent frame and the force-sensor electronics are placed outside the high-field region or coupled through a nonmagnetic force-transfer member.

An independently supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly produces the static field, preferably 1500–2500 Oe for the reference sphere, with magnitude and polarity under controller command. The reference source is a 76 mm-pole C-frame electromagnet of the GMW-5403 class, operated at approximately 20 A and an approximately 80 mm pole gap to provide about 0.196 T according to its characterized magnet curve. The magnet and cavity share a rigid source frame mechanically independent of the sample force flexure. At the reference continuous air-cooled operating point the magnet is a roughly 0.20-kW thermal source (approximately 20 A and 10 V), so coil and yoke temperatures are logged and field-sense reversals are performed in separate thermally settled blocks. The field and gradient maps are repeated after each reversal; fast randomized drive and control states are

executed within a fixed-bias block. Shaped pole faces, trim windings, shim elements, or a combination thereof provide the required field uniformity. A 15 cm-diameter, 200-turn-per-coil air-core Helmholtz pair at 5 A produces only about 0.012 T (approximately 120 Oe), and therefore is not the reference source for a 2000 Oe operating point; any air-core alternative must disclose and validate its current, turn count, conductor heating, duty cycle, and measured field map.

Before a force run, the vertical force-axis component of the field gradient and the other first spatial derivatives are mapped through the sample and sensor travel volumes with calibrated probes and verified by translating a test moment. For the reference saturated sphere, a nominal room-temperature saturation induction of approximately 0.178 T (the customary YIG value stated as $4\pi M_s$ approximately 1780 G) gives a magnetic moment of approximately $0.074 \text{ A} \cdot \text{m}^2$; the actual magnetization curve is measured for the specimen used. Therefore a 0.01 mg-force-equivalent threshold (approximately 98 nN) can be imitated by a force-axis gradient of only approximately $1.32 \text{ } \mu\text{T/m}$. Vendor field-uniformity specifications do not certify this force threshold. The apparatus therefore uses gradient shims or controlled gradient dither and extrapolates the registered force contrast to zero gradient. The residual uncertainty in the force-axis gradient component is required to remain below a preregistered fraction of the $1.32\text{-}\mu\text{T/m}$ gradient threshold, preferably one third (approximately $0.44 \text{ } \mu\text{T/m}$ for the stated example), or the run is invalid. For the FIG. 8 axis convention, the complete magnetic-force account uses the vertical component of $\nabla(\mu \cdot B)$, including $\mu_z \partial B_z / \partial y$, rather than assuming that one coordinate derivative alone governs. Field reversal is a settled test block, not by itself proof of a nonmagnetic force.

The drive chain comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), pulse modulator, power amplifier (reference 1–5 W for commissioning), directional coupler, and incident/reflected/transmitted-power detectors. Pulse width may lie from 0.1–10 μs and repetition rate from 50 Hz–1 MHz only in combinations satisfying pulse nonoverlap and a controller-enforced duty-factor limit; the reference commissioning range is 0.001–0.10 duty. Higher apparent cross-products are implemented as finite bursts separated by recovery intervals, not overlapping pulses. Loaded-Q, cavity ring-up/ring-down, coupling, mode volume, and peak and average absorbed power are measured; $Q \times \text{source power}$ is not treated as a power or field multiplier.

The controller records the drive-state vector and synchronously logs: sample force; source-frame reaction force; optional outer-platform support force; incident, reflected, and transmitted power; cavity, sample, stem, coil, sensor, and ambient temperature; pressure; three-axis magnetic field; vibration and acceleration; strain at the force path and feed supports; and source/reference timing. A calorimetric channel or measured temperature-history control estimates absorbed energy. The acceleration-to-apparent-force transfer function is calibrated over the registered force bandwidth. For the 2.71-g reference body, 98 nN corresponds to approximately $3.62 \times 10^{-5} \text{ m/s}^2$; the preferred one-third nuisance allowance corresponds to approximately $1.21 \times 10^{-5} \text{ m/s}^2$ before the measured mechanical transfer is applied. The residual acceleration- coupled force contribution, rather than accelerometer resolution alone, must remain below the preregistered nuisance allowance. The population coherence time is measured on

the RF/magnetic channel; the force channel records only dynamics within its own measured bandwidth. A 1 kHz force channel therefore tests millisecond-class envelopes and does not purport to resolve a 100 ns material decay.

Controller sequence and control matrix. Before collecting the locked confirmation set, the operator may run an exploratory resonance, power, pulse, polarization, orientation, and repetition-rate survey. The search space and selected operating points are then frozen. Opposite alignment-field senses are established in separate thermally settled bias blocks, with field and gradient remapped after each reversal. Within each fixed-bias block the controller randomizes, and may blind the analyst to, repeated blocks containing: field-off; field-on/drive-off; resonant drive; detuned drive adjusted to match absorbed energy and temperature history; active YIG; a geometry-, mass-, thermal-, dielectric-loss-, and mechanical-response-matched reference; opposite circular drive polarizations where supported; sample and reference swaps; force-sensor or apparatus-position swaps; and positive-control force injections of known magnitude and phase. A magnetic reference with characterized susceptibility and magnetostriction is included where needed; a non-alignable mass alone is not treated as a complete control.

The analysis returns one of three classifications: (i) null within the registered uncertainty; (ii) response accounted for by one or more measured nuisance channels; or (iii) qualifying unexplained drive-correlated response. It does not label class (iii) a validated change in measured weight until all gates pass:

- **A1 — repeatability:** the registered ON/OFF response repeats across blocks and days;
- **A2 — resonance contrast:** the response differs from an absorbed-energy-matched detuned state by the preregistered amount;
- **A3 — material contrast:** it survives active/reference swaps and is not reproduced by the matched reference;
- **A4 — magnetic-force exclusion:** the mapped force-axis component of $\nabla(\mu \cdot B)$, including measured field and moment changes, cannot account for the response within uncertainty;
- **A5 — thermal/radiometric exclusion:** calorimetry, temperatures, pressure sweeps, and thermal-history-matched shams cannot account for it;
- **A6 — mechanical/electrical exclusion:** vibration, acceleration, strain, cable/feed forces, ground/reference coupling, and local reaction force cannot account for it;
- **A7 — orientation and rotating-frame matrix:** any field-sense, polarization, sample-orientation, apparatus-rotation-sense, and local-frame-angular-rate parity is learned consistently from held-out data rather than imposed after inspection; a rotation control records angular rate, angular acceleration, rotor or sample orientation, and response latency, and compares centered frame-only rotation with displaced turntable rotation;
- **A8 — statistical validity:** effect size exceeds the preregistered uncertainty threshold after correction for the exploratory search and multiple comparisons; balance readability is not treated as standard deviation; and

- **A9 — independent confirmation:** a second apparatus, sample, run series, or laboratory reproduces the registered sign and response surface within stated uncertainty.

Only an A1–A9 result is a validated support-force response under Definition 18. A null result is a complete and useful operation of the claimed qualification method; the unit is then a null reference (Definition 23).

E2 — Prophetic operating species: contained unloading, station-keeping, and soft-landing qualification (conditional on a qualified module class). If qualification (E1) establishes a qualified responsive module class whose validated map extends toward a substantial fraction of the body's baseline support force, a scale-up qualification apparatus tests whether the measured response characteristic reaches a support-force null. No null crossing, linear power law, or required input power is presumed. The apparatus adds a vertical drive-transparent containment guide, optical position sensing, high-speed video, an extended uniform-field region, and one or more segmented or traveling coupling structures whose overlap is measured through the travel range. Drive authority is increased only in steps within the validated thermal, magnetic, and response envelope. A position controller may be enabled only after the measured response map supplies sign, gain, bandwidth, saturation, hysteresis, and stability margins. The sequence is: system check; artifact-control replay; calibration at the lower-power validated points; held-out interpolation tests; incremental power extension; and, only if the measured force approaches zero without a failed gate, contained lift and drive-kill tests. Safety interlocks include drive leakage, coil and sample temperature, guide position, loss of resonance, unexpected reaction force, and containment breach. Operating species that this qualification would enable—station-keeping of a body at a commanded position, soft landing of a descending body, and incremental unloading of a supported load—are performed by the deployed operating method of FIG. 2 with a position or descent-rate input to the setpoint controller, and only where the class map covers the required force fraction. Within the applicant's hypothesis, a confirmed unsupported stationary state would be described as radius command; the disclosure does not represent that state as achieved.

E3 — Plural-module support-force system (600) and array qualification (FIG. 6). A plural-module support-force system comprises a load-bearing structure (602) supporting a load; a plurality of calibrated support-force modules (300₁ ... 300_n) of E5, each holding its own empirical response map, mounted so that each body's support force is transmitted through its support interface to the structure while its reaction frame loads pass through a separate reaction path; a common phase reference (610) with per-module amplitude and phase trim; a system-level force account (620) comprising at least one force sensor reading the support force of the structure or a calibrated sum of all instrumented support and reaction paths; a system calibration memory (630) holding the array allocation map (Definition 28); and a system controller (640) that receives a system support-force setpoint, allocates per-module setpoints from the array allocation map within each module's permissible interpolation region, monitors per-module validity states and the system-level account, and transitions every module to safe-off when the system-level

account deviates from the allocation map beyond a stored tolerance or any module reports invalid. Differential per-module setpoints for attitude or trim of the structure are commanded only from differential responses recorded in the allocation map.

Array qualification. The assembled plurality is driven from the common phase reference through per-module amplitude and phase trim. Each module has a measured resonance, absorbed power, field map, temperature, and preferably its own force channel; the complete assembly also has a force and reaction-force channel. The qualification controller compares one-module, sequential, in-phase, randomized-phase, and inactive-module states. Additivity, cancellation, coupling-mode splitting, and thermal and magnetic cross-talk are measured outputs. If a nonzero response is shown to add within uncertainty, the measured array law—not an assumed population count—is written to the array allocation map and used for engineering scale-up; if it does not add, the measured allocation is written as it stands. In a lift or unloading configuration the system adds the containment guide and position sensing of E2, and incremental unloading is enabled only when every operated module is qualified, the allocation map covers the commanded fraction, and the system-level measured-force, reaction-force, stability, and energy accounts satisfy preregistered limits.

E4 — Materials envelope and candidate-screening protocol. The enabled reference species is a room-temperature single-crystal YIG sphere under a measured saturation field and selected FMR mode. Other iron garnets, including Ga-, Al-, La-, or Bi-substituted compositions, are candidate species only after their saturation, resonance spectrum, linewidth, phase response, dielectric loss, magnetostriction, thermal behavior, and force-control residuals are measured under the E1 protocol. Single-crystal spheres are preferred for field uniformity and a tractable mode spectrum; films, polycrystals, and other geometries are separately characterized rather than presumed equivalent.

Cryogenic operation is an optional candidate condition. The selected material's linewidth and relaxation are measured at each temperature; cooling is not presumed universally to narrow the line. Doped corundum, olivine-class crystals, dynamic nuclear-orientation media, parametrically pumped modes, and layered, doped, strained, composite, or patterned media are prophetic screening branches. For each branch, the alignment mechanism, composition, temperature, resonance or pump condition, coherence observable, absorbed-power path, and artifact controls are established before a force result is classified. Meeting $\tau_v \geq 1$ alone does not establish a support-force response.

E5 — Calibrated support-force module (300) (FIG. 5). A sealed module comprises the body of ordered medium in its holder (310), the coupling structure and locked drive chain (330), the alignment-field source with shims or trim winding (320), an embedded controller (340) executing the deployed operating method, a calibration memory (345) holding the module's empirical response map and provenance, witness sensing (350) comprising the runtime witnesses and, optionally, a support-path force sensor, and a standard interface (360): a mounting interface transmitting the body's support force to the external structure while reaction loads pass to the module frame and thence by a separate path; an electrical interface for drive and field power; and a data interface that accepts a support-force setpoint or test-state command and reports achieved change, uncertainty, validity state, lock state, and health. The controller

enables setpoint mode only when the map contains at least one independently validated nonzero operating point; a module whose map contains only null points operates as a null reference, accepts test-state commands, and reports L0 and L1 quantities. The module carries a unit identifier bound to its provenance record and refuses a map that does not match it. Runtime-reversible and fixed-bias configurations are disclosed; the sign associated with either configuration is determined by the map, not presumed from field polarity. Modules are qualified individually or accepted under a production-class map after the acceptance test of the qualification section.

E6 — Prophetic operating species (conditional on a qualified module class and within its map).

Each of the following is an operating mode of the deployed operating method (FIG. 2) executed by a qualified responsive module or plural-module system, limited to the measured response, bandwidth, energy, thermal, orientation, and stability envelope of its map; none is represented as achieved, and none is available to a null reference:

- *Payload trim* — a setpoint expressed as a commanded fraction of the baseline support force of a load rigidly borne with the module, with sign selected from the map; the achieved fraction is reported with uncertainty.
- *Time-varying support-force profiles* — steps, ramps, sinusoids, and user-defined profiles within the map's bandwidth, slew, and distortion limits.
- *Station-keeping* — holding a body at a commanded position against a position or descent-rate input, only where the class map covers the required force fraction (E2).
- *Soft landing* — a descent-rate-referenced profile reducing the support-force deficit of a descending body, under the same condition.
- *Measured array allocation* — distribution of a system setpoint across modules from the array allocation map, including differential allocation for attitude or trim (E3).
- *Balancing or trimming of rotating or suspended structures* — differential or time-varying setpoints applied by modules mounted on the structure.

PROPHETIC EXAMPLES (paper examples; no experiment, nonzero result, or qualified unit represented as achieved)

P1 — commissioning and sensitivity. A 10.00 mm YIG sphere has nominal volume 0.5236 cm³ and nominal mass approximately 2.71 g; actual dimensions and mass are measured. It is characterized near a selected FMR mode under approximately 2000 Oe and driven at a reference 3 W pulse-peak setting, 2 μs width, and 10 kHz repetition (duty 0.02; incident time-average 0.06 W before coupling loss). A 0.01 mg-force-equivalent balance increment is approximately 98 nN, or 3.7 ppm of the sphere's ordinary supported weight. That number is an instrument readability target, not a standard deviation and not a predicted response. Baseline noise, Allan deviation, drift, effective independent sample count, and the locked analysis determine the actual detection limit. The completed experiment reports one of the classification levels in E1, including a null or upper bound.

P2 — artifact challenge. With nominal saturated moment approximately $0.074 \text{ A} \cdot \text{m}^2$, a 98 nN magnetic force corresponds to approximately $1.3 \text{ } \mu\text{T/m}$ if the entire moment changes; for a fractional drive-induced moment change ϵ , the corresponding gradient is approximately $(1.3/\epsilon) \text{ } \mu\text{T/m}$. The apparatus maps and dithers the gradient, measures ΔM where practicable, and extrapolates the registered drive contrast to zero gradient. Resonant and detuned or phase-scrambled states are matched by absorbed-energy and temperature history. A candidate that fails gradient, reaction-force, thermal, position-swap, or blinded-holdout gates is classified as an artifact even if synchronous with resonance or field reversal.

P3 — qualification of a unit and transfer of its map to deployment. If P1/P2 and an independent replication establish a validated support-force response under Definition 18 and gates A1–A9 at one or more operating points, the qualification controller estimates R on a multidimensional grid of drive states, reserves held-out points for validation, and writes to the unit's calibration memory the validated points with uncertainty, the permissible interpolation region bounded by those points, the runtime limits observed during qualification, the safe-off parameters, and the provenance record; the unit ships as a qualified responsive module. Its deployed controller accepts setpoints only inside that region. If no operating point passes, the same memory records a null map and the unit ships as a null reference. No full-weight-null power, percentage trim, slew time, array law, or conversion efficiency is predicted in this disclosure; those quantities are outputs of the qualification and scale-up program.

P4 — conditional lift and array branch. If a validated response remains inside its measured interpolation envelope at a substantial fraction of supported weight and the system-level force, reaction, stability, and energy accounts close at that operating point, E2 tests incremental unloading in containment. If individually validated module responses are reproducible, E3 array qualification tests N -module scaling under sequential, coherent, phase-scrambled, and inactive-module controls and writes the measured allocation. Unsupported station-keeping, payload trim, and lift remain expressly prophetic until those conditions and gates A1–A9 pass at the proposed scale.

P5 — deployed payload-trim operation of a qualified module (prospective). A qualified responsive module built on the 2.71-g reference body (baseline support force approximately 26.6 mN) is mounted rigidly under a payload deck in the mounting class and orientation recorded in its map. Its class map is assumed, for this paper example only, to contain at least one validated nonzero operating point $\Delta F_v \pm \sigma_v$ at drive state x_v ; no value of ΔF_v is predicted. An operator commands a payload-trim setpoint ΔF_{set} . The controller accepts the command only if $|\Delta F_{\text{set}}| \leq |\Delta F_v|$ and the interpolated state lies within the permissible region; ramps the field (if the map's sign selection requires a sense change) in a settled block; applies the locked drive at the interpolated state; reports the achieved change as $\Delta F_{\text{set}} \pm \sigma(x)$, with $\sigma(x)$ propagated from the map covariance; interleaves drive-off zero checks at the stored interval; and transitions to safe-off if a runtime witness leaves its limit. The smallest reportable setpoint is bounded below by $\sigma(x)$, not by the 98-nN (3.7 ppm of the body's baseline) qualification-instrument readability of P1. The trimmed fraction of the payload's own support force equals ΔF_{set} divided by the payload's baseline support force and is a reported, not presumed, quantity.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

The empirical-response-map architecture above is a preferred qualification and production species, not the sole statement of the invention. In the POAMS-native embodiment, resonant phase-coherent drive reallocates the body's angular-momentum ledger between intrinsic turn and orbital winding; the resulting natural-orbit-radius change appears at a support as a changed constraint bill. This mechanism and its sign-selection sequence are disclosed prophetically and are not withdrawn merely because extant force ontology supplies no transfer law. A unit may therefore be built and operated directly under the stated alignment, resonance, sense, and support-interface sequence while its achieved magnitude is measured; no fixed magnitude, efficiency, or successful result is represented as achieved.

CLAIMS

1. A support-force control apparatus comprising: a body comprising an ordered medium having an internal angular-momentum population alignable by a static field; a support interface through which a support force of the body is transmitted to a supported or supporting structure; an alignment-field source configured to apply to the body an alignment field of a magnitude and sense establishing a measured alignment state of the population; a coupling structure and a drive source configured to apply to the body an oscillating drive at a located precession resonance of the aligned population; a reaction frame carrying the alignment-field source and the coupling structure and transmitting their support loads through a reaction load path separate from the support interface; a calibration memory storing an empirical response map for the apparatus or for a production class to which the apparatus belongs, the empirical response map relating a drive state of the apparatus to a change in the support force of the body, having been established by measurement of the body or of the production class, and containing at least one independently validated nonzero operating point, an uncertainty for each operating point, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; runtime witness sensors configured to measure at least a temperature of the body, an orientation, a force-axis magnetic-field-gradient witness, an acceleration, and an absorbed drive power; and a support-force setpoint controller configured to receive a support-force setpoint, to determine from the empirical response map a drive state corresponding to the setpoint when the setpoint lies within the permissible interpolation region and to inhibit the setpoint otherwise, to command the alignment-field source and the drive source to the determined drive state, to compare outputs of the runtime witness sensors with the runtime limits during operation, to report an achieved change in the support force with a propagated uncertainty, and to transition the apparatus to a safe-off state defined by the safe-off parameters and to set a recalibration flag when a runtime limit is exceeded or a stored validity condition of the empirical response map fails.
2. The apparatus of claim 1, further comprising a force sensor in the support interface producing a measured support-force signal, wherein the support-force setpoint controller is configured to combine feedforward from the empirical response map with feedback on the measured support-

force signal, with loop gains bounded by a bandwidth, a hysteresis, and time constants stored in the empirical response map.

3. The apparatus of claim 1, wherein the support interface is free of a force sensor and the support-force setpoint controller operates in feedforward from the empirical response map with the runtime witness sensors and an interleaved drive-off zero check as the stored validity conditions.
4. The apparatus of claim 1, wherein the alignment-field source comprises a permanent-magnet yoke with a trim winding or an electromagnet of commanded sense, the empirical response map stores the change in the support force at each sense, and the support-force setpoint controller selects the sense for a commanded change from the empirical response map.
5. The apparatus of claim 1, wherein the coupling structure comprises a TE-mode cavity, loop-gap resonator, re-entrant resonator, waveguide, or coil structure placing the body at a mapped microwave magnetic-field antinode transverse to the alignment field, or comprises another resonator having a mapped off-axis location with such a field.
6. The apparatus of claim 1, wherein the drive source comprises a pulse or burst modulator enforcing a programmed duty-factor bound with nonoverlapping pulses, and the support-force setpoint controller limits absorbed drive energy to a thermal envelope stored in the empirical response map, higher delivered energy being implemented as bursts separated by recovery intervals or as thermally matched continuous-wave operation.
7. The apparatus of claim 1, further comprising a reaction-force sensor on the reaction load path, wherein the support-force setpoint controller is configured to report the achieved change as invalid when the achieved change is accounted for within a stored tolerance by an equal and opposite change measured by the reaction-force sensor.
8. The apparatus of claim 1, wherein the empirical response map further stores a hysteresis with drive history, thermal coefficients, and orientation coefficients of the change in the support force, and the support-force setpoint controller compensates the determined drive state therefor or restricts the permissible interpolation region accordingly.
9. The apparatus of claim 1, wherein the body is a room-temperature single-crystal yttrium-iron-garnet sphere having a diameter between 2 mm and 15 mm, or a substituted iron garnet or engineered medium whose population coherence time τ and drive turn rate ν satisfy $\tau \cdot \nu \geq 1$ and whose empirical response map was established by measurement of that medium.
10. The apparatus of claim 1, wherein the support-force setpoint controller reports a validity state selected from valid, degraded, out-of-envelope, invalid, and safe-off, permits setpoint operation only in the valid state, and executes drive-off zero checks at an interval and a recalibration schedule stored in the calibration memory.
11. The apparatus of claim 1, wherein the support-force setpoint comprises a time-varying profile of support-force change, and the support-force setpoint controller limits the profile to a bandwidth, slew rate, and distortion envelope stored in the empirical response map.

12. A method of operating a support-force control apparatus having a body of ordered medium on a support interface, an alignment-field source, a coupling structure and drive source, runtime witness sensors, and a calibration memory storing an empirical response map that relates a drive state to a change in a support force of the body, the empirical response map having been established by a qualification procedure performed before the method and containing at least one independently validated nonzero operating point, an uncertainty for each operating point, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, the method comprising: loading the empirical response map and verifying that the provenance record matches an identifier of the apparatus or of its production class; executing a drive-off zero check and comparing its result with a stored baseline; receiving a support-force setpoint; determining whether the setpoint lies within the permissible interpolation region and inhibiting the setpoint when it does not; determining from the empirical response map a drive state corresponding to the setpoint; applying the alignment field and the oscillating drive at a located precession resonance of the body at the determined drive state; regulating the change in the support force by feedforward from the empirical response map, by feedback on a measured support force, or both; comparing outputs of the runtime witness sensors with the runtime limits during the applying; reporting an achieved change in the support force with a propagated uncertainty; and transitioning the apparatus to a safe-off state defined by the safe-off parameters and setting a recalibration flag when a runtime limit is exceeded or a stored validity condition fails; wherein the qualification procedure is not repeated during the method.
13. The method of claim 12, further comprising interleaving drive-off zero checks at an interval stored in the calibration memory and, when a drift determined from the zero checks exceeds a stored tolerance, reporting a degraded state and shrinking the permissible interpolation region or setting the recalibration flag.
14. The method of claim 12, wherein the support-force setpoint is a commanded fraction of a baseline support force of a load rigidly borne with the body, and a sign of the commanded fraction is selected from the empirical response map.
15. The method of claim 12, wherein the support-force setpoint is a time-varying profile and the applying is limited to a bandwidth, slew rate, and distortion envelope stored in the empirical response map.
16. The method of claim 12, further comprising recording incident, reflected, transmitted, absorbed, dissipated, mechanical, thermal, and stored energy and a reaction-frame force for the determined drive state, and reporting a bounded energy and momentum budget for the achieved change.
17. The method of claim 12, further comprising receiving a position or descent-rate signal for the body or for a structure borne with it and adjusting the support-force setpoint from that signal, the adjusting being permitted only while the adjusted setpoint remains within the permissible interpolation region and otherwise inhibited.

18. The method of claim 12, further comprising, after the recalibration flag is set, performing a reduced re-qualification comprising the drive-off zero check and a check of at least one validated operating point against a separate force reference, and disabling setpoint operation and returning the apparatus to the qualification procedure when the reduced re-qualification fails.
19. A calibrated support-force module comprising: a sealed housing; a body comprising an ordered medium having an internal angular-momentum population alignable by a static field; an alignment-field source; a coupling structure and drive source configured to apply to the body an oscillating drive at a located precession resonance of the aligned population; a mounting interface transmitting a support force of the body to a structure external to the housing, reaction loads of the alignment-field source and the coupling structure being transmitted through a separate load path; runtime witness sensors; a calibration memory storing an empirical response map relating a drive state to a change in the support force of the body, established by measurement of the body or of a production class of the module, with an uncertainty for each operating point, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; an embedded controller; and a data interface configured to accept a support-force setpoint or a test-state command and to report an achieved change in the support force, its uncertainty, a validity state, a lock state, and a health state; wherein the embedded controller enables a setpoint mode, in which the module commands a drive state determined from the empirical response map for a received setpoint within the permissible interpolation region, only when the empirical response map contains at least one independently validated nonzero operating point, and otherwise operates the module as a null reference that executes test-state commands and reports the change in the support force synchronized with each commanded state.
20. The module of claim 19, wherein the calibration memory stores a unit or class identifier bound to the provenance record and the embedded controller refuses an empirical response map whose provenance record does not match the identifier.
21. The module of claim 19, wherein the mounting interface and the data interface conform to a standard interface for integration of a plurality of the modules into a plural-module system, and the data interface further reports the permissible interpolation region and the runtime limits of the module to a system controller.
22. The module of claim 19, wherein the alignment-field source is a fixed-bias permanent-magnet yoke and the sign of the change in the support force available in the setpoint mode is the sign recorded in the empirical response map for that fixed bias, or the alignment-field source comprises a trim winding by which the embedded controller selects a sense recorded in the empirical response map.
23. A plural-module support-force system comprising: a load-bearing structure supporting a load; a plurality of support-force modules, each comprising a body of ordered medium, an alignment-field source, a coupling structure and drive source configured to apply an oscillating drive at a located precession resonance of the body, a mounting interface transmitting a support force of the body to

the load-bearing structure, and a calibration memory storing an empirical response map of that module relating a drive state to a change in the support force of its body, established by measurement, and containing at least one independently validated nonzero operating point, an uncertainty, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters; a system-level force account comprising at least one force sensor reading a support force of the load-bearing structure or a calibrated sum of instrumented support paths and reaction paths; a system calibration memory storing an array allocation map established by measurement of the assembled plurality and relating sets of per-module drive states to a change in the support force of the load-bearing structure with uncertainty; and a system controller configured to receive a system support-force setpoint, to allocate per-module setpoints from the array allocation map within the permissible interpolation region of each module, to monitor a validity state of each module and the system-level force account, and to transition every module to a safe-off state when the system-level force account deviates from the array allocation map beyond a stored tolerance or any module reports an invalid state.

24. The system of claim 23, further comprising a common phase reference with per-module amplitude and phase trim, wherein the array allocation map stores responses measured under in-phase, phase-scrambled, and sequential operation of the modules and cross-talk terms between modules.
25. The system of claim 23, wherein the system controller commands differential per-module setpoints for attitude or trim of the load-bearing structure only from differential responses recorded in the array allocation map.
26. The system of claim 23, further comprising reaction-force sensors on the reaction paths of the modules, wherein the system controller reports the change in the support force of the load-bearing structure as invalid when it is accounted for within a stored tolerance by measured reaction forces.
27. The system of claim 23, wherein the array allocation map was established by an array qualification comprising operating the modules sequentially, simultaneously in phase, simultaneously with phase-scrambled references, and with selected modules inactive, and comparing the measured change in the support force of the load-bearing structure with the per-module empirical response maps without presuming additivity.
28. The system of claim 23, further comprising a containment guide and position sensing for the load-bearing structure, wherein the system controller enables incremental unloading of the load-bearing structure only when every operated module holds an empirical response map with at least one independently validated nonzero operating point, the array allocation map covers the commanded fraction of the support force, and system-level measured-force, reaction-force, stability, and energy accounts satisfy preregistered limits.
29. A method of qualifying a support-force module and establishing its empirical response map, comprising: mounting a body comprising an ordered medium on a sample force sensor; supporting an alignment-field source and an oscillating-drive coupling structure independently of the sample force sensor; measuring a static-field alignment state of the body; locating a precession resonance

of the body; applying an oscillating drive at and away from the resonance in a randomized sequence of drive and control states; recording support force, reaction force on the independently supported alignment-field source or coupling structure, absorbed drive energy, temperature, magnetic-field gradient, vibration, acceleration, pressure, strain, and reflected or transmitted drive power; fitting nuisance responses to the recorded measurements; estimating an empirical response characteristic relating the drive state to a change in the support force with uncertainty; classifying each operating point as a null response, an identified artifact, or an unexplained drive-correlated residual according to predetermined acceptance gates; validating any nonzero residual by independent replication; and writing to a calibration memory of the module an empirical response map comprising the validated operating points and their uncertainties, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, together with a qualification status of qualified responsive module when the empirical response map contains at least one independently validated nonzero operating point and of null reference otherwise.

30. The method of claim 29, wherein the oscillating drive is pulsed with nonoverlapping pulses, and pulse width and repetition rate are selected subject to a duty-factor bound, with a higher-duty condition identified and thermally matched as a burst or continuous-wave condition rather than as a train of discrete pulses.
31. The method of claim 29, further comprising servo-locking the drive frequency to the located precession resonance and recording the identified precession mode, its spatial field pattern and linewidth, the loaded quality factor of the coupling structure, and coupling-structure energy ring-up and ring-down times before a support-force residual is evaluated.
32. The method of claim 29, further comprising mapping or dithering the static-field gradient at the body, fitting or extrapolating the measured change in the support force to a zero-gradient condition, and invalidating an operating point when the residual uncertainty in the force-axis gradient exceeds a preregistered fraction of the gradient that would imitate the smallest reportable change acting on the measured magnetic moment of the body.
33. The method of claim 29, further comprising repeating randomized observation blocks with the body, a substantially nonmagnetic reference body, and a magnetic reference body selected to challenge field-gradient, susceptibility, heating, strain, and vibration artifacts, and swapping positions of the body and at least one reference body, swapping force-sensor positions, or both, while preserving the randomized observation blocks.
34. The method of claim 29, further comprising comparing the recorded support force with the recorded reaction force and, where all external support paths are instrumented, computing their calibrated sum, and classifying as an identified artifact any change in the support force accounted for by the reaction force.
35. The method of claim 29, wherein the randomized sequence includes resonant drive, static-field-only, drive-only, reversal of alignment-field sense, polarization-reversal, and drive-off states and at least one absorbed-energy-matched control comprising detuned, phase-scrambled, or sham drive

matched to the resonant state by measured absorbed energy and temperature history rather than by equal source power alone.

36. The method of claim 29, wherein an exploratory scan is separated from a frozen, preregistered confirmation run using blinded state labels, held-out operating points, and multiple-comparison control, and the predetermined acceptance gates comprise repeatability across blocks and days, resonance contrast against the absorbed-energy-matched control, material contrast against reference bodies, exclusion of a force imitated by the mapped field gradient acting on the measured magnetic moment, thermal and radiometric exclusion, mechanical and electrical exclusion including reaction force, an orientation and rotating-frame parity learned from held-out data, statistical validity after correction for the exploratory search, and independent confirmation on a second apparatus, body, run series, or laboratory.
37. The method of claim 29, further comprising a pressure sweep that measures or bounds convection, radiometric force, outgassing, and pressure-dependent thermal transport.
38. The method of claim 29, further comprising qualifying a plurality of modules of common design, establishing a production-class empirical response map from their validated operating points, and accepting an additional module of the design under the production-class map only after an acceptance test at one or more validated operating points against a separate force reference agrees with the production-class map within a stored tolerance.
39. A qualification apparatus for establishing the empirical response map of a support-force module, comprising: a body comprising an ordered medium; a sample force sensor supporting the body; an alignment-field source configured to apply an alignment field to the body and supported independently of the sample force sensor; an oscillating-drive coupling structure configured to couple an oscillating drive to the body and supported independently of the sample force sensor; a drive source configured to excite an identified precession mode of the body; a reaction-force sensor supporting at least part of the alignment-field source, the coupling structure, or a mechanically independent source frame carrying one or both; a three-axis field sensor and field-gradient mapping probe, thermal sensor, vibration sensor, acceleration sensor, pressure sensor, strain sensor, and drive-power monitor; a controller configured to execute randomized drive and control states, to estimate an empirical response characteristic relating drive state to a change in the support force with uncertainty, and to classify each operating point as a null response, an identified artifact, or an unexplained residual according to predetermined acceptance gates; and a calibration-memory writer configured to write to a calibration memory of the module an empirical response map comprising validated operating points, uncertainties, a provenance record, a permissible interpolation region, runtime limits, and safe-off parameters, and a qualification status of qualified responsive module or null reference.
40. The qualification apparatus of claim 39, wherein the coupling structure comprises a TE-mode cavity, loop-gap resonator, re-entrant resonator, waveguide, or coil structure and places the body at a mapped microwave magnetic-field antinode transverse to the alignment field, or comprises

another resonator having a mapped off-axis location with such a field.

41. The qualification apparatus of claim 39, wherein the drive source comprises a pulse or burst modulator that enforces a programmed duty-factor bound, and wherein the controller records incident and reflected energy for each state window and, for a two-port coupling structure, transmitted energy.
42. The qualification apparatus of claim 39, wherein the alignment-field source comprises an independently supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly having shaped pole faces or shim elements, together with probes for mapping field magnitude and gradient across the body, and the controller performs alignment-sense reversals in separate thermally settled blocks with the field and gradient remapped after each reversal.
43. The qualification apparatus of claim 39, further comprising an outer support-force sensor configured to measure the aggregate external support force of the body and the mechanically independent source frame.
44. The qualification apparatus of claim 39, further comprising a calibrated magnetic-force injector, heater, mechanical actuator, or electrical reference injector configured to validate the acquisition and artifact-classification chain.
45. The qualification apparatus of claim 39, wherein the body is a room-temperature single-crystal yttrium-iron-garnet sphere having a diameter between 2 mm and 15 mm, and the controller records its measured mass, magnetization curve, saturation state, resonant mode, and linewidth.
46. The qualification apparatus of claim 39, wherein the controller compares measurements obtained at a room-temperature reference state and at least one additional-temperature state only after the mode linewidth, thermal time constants, magnetic state, and force-sensor response have been measured at each compared temperature.
47. The qualification apparatus of claim 39, configured as an array qualification fixture further comprising a plurality of support-force modules mounted to a common load-bearing structure, a common phase reference with per-module amplitude and phase trim, a force channel for each module, and a force channel and reaction-force channel for the assembled structure, wherein the controller operates the modules sequentially, simultaneously in phase, simultaneously with phase-scrambled references, and with selected modules inactive, and the calibration-memory writer writes an array allocation map relating sets of per-module drive states to the measured change in the support force of the structure without presuming additivity.
48. The qualification apparatus of claim 39, further comprising a rotation stage or rotatable constraint frame, an angular-position encoder, and an angular-rate or inertial sensor, the controller being configured to compare measurements at opposite rotation senses and at a plurality of angular rates, to record angular acceleration and response latency, and to distinguish rotation of a centered local constraint frame from rotation of the body at a nonzero displacement from the rotation-stage center.

49. A prophetic POAMS-native support-force control apparatus comprising: a body of ordered medium having an internal angular-momentum population; an alignment-field source setting an alignment sense; a coupling structure and drive source exciting a phase-coherent precession mode; a support interface transmitting the body's constraint force; and a controller configured to select an intrinsic-turn or orbital-winding allocation by drive sense, phase, polarization, or mode and thereby command a natural-orbit-radius change read at the support interface as a support-force change, an achieved magnitude being an output of operation and not a fixed limitation of the apparatus.
50. A method of operating the apparatus of claim 49, comprising: establishing the alignment sense; locating and locking the precession resonance; applying a phase-coherent drive in a selected sense or mode; reading the support force relative to a drive-off state; reversing at least one of alignment sense, drive sense, polarization, or mode to test the predicted channel sign; and reporting the measured support-force change or a null result without requiring a pre-existing empirical response map.

ABSTRACT

A support-force control apparatus comprises a body of ordered medium on a support interface, an alignment-field source, a coupling structure and locked drive at the body's precession resonance, a separately loaded reaction frame, a calibration memory holding an empirically established response map with at least one independently validated nonzero operating point, uncertainty, provenance, permissible interpolation region, runtime limits, and safe-off parameters, and a setpoint controller that commands a selected change in support force from the map, regulates by feedback or feedforward, monitors runtime witnesses, and enters safe-off on a validity failure. Calibrated modules, plural-module systems using per-module maps and a measured array allocation, and deployed operating methods are disclosed. The map is established once per unit or class by a null-valid qualification apparatus and protocol with randomized controls, gradient mapping, reaction-force accounting, and acceptance gates; a null map yields a null reference. No sign, magnitude, additivity, or scaling is presumed.

Force-free-radius and measured-weight control apparatus (400)

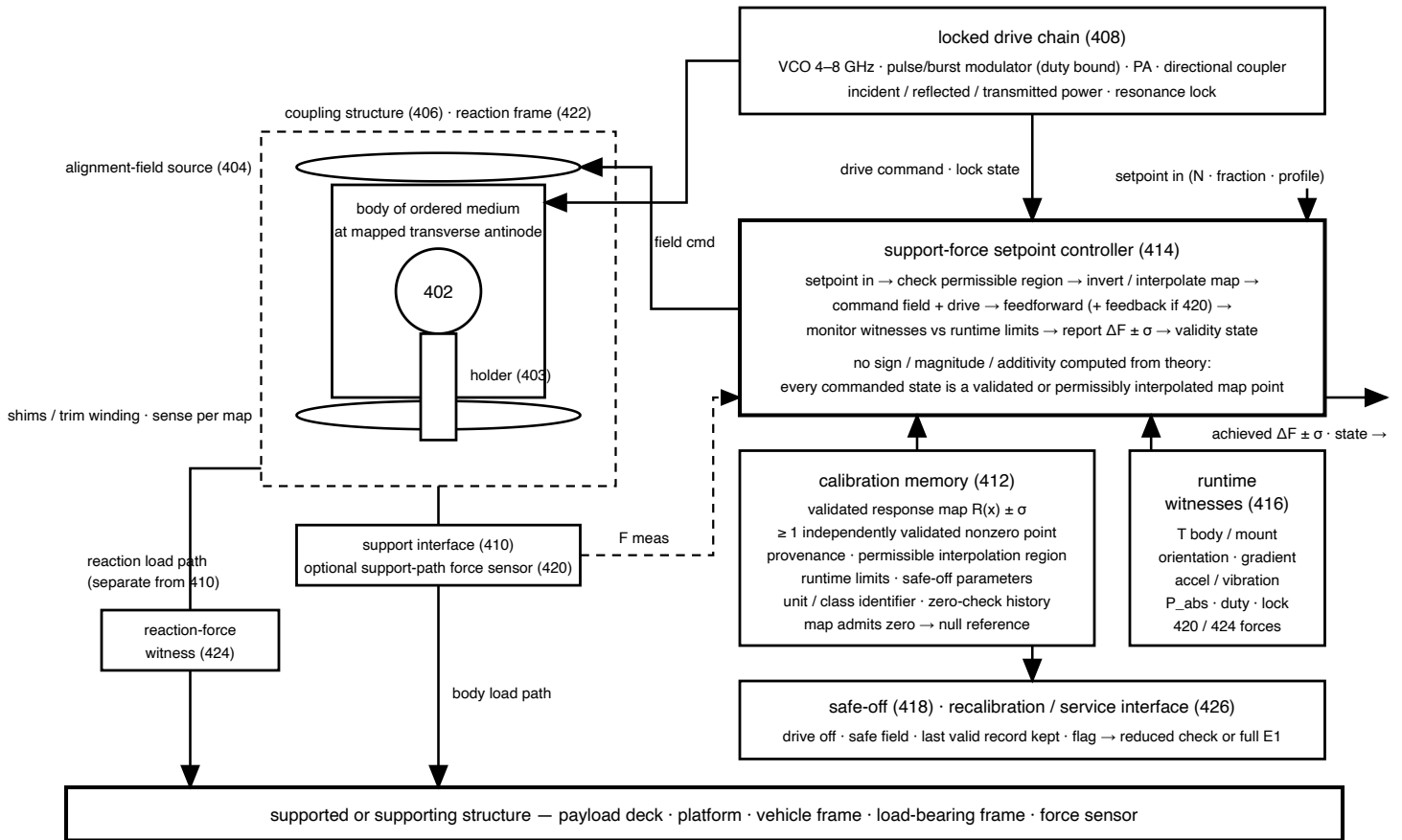


FIG. 1

Force-free-radius and measured-weight control apparatus (400) — body of ordered medium (402) in a holder (403) on a support interface (410) to the supported or supporting structure; alignment-field source (404) and coupling structure (406) carried by a reaction frame (422) on a separate reaction load path with a reaction-force witness (424); locked drive chain (408); calibration memory (412) holding the validated response map; support-force setpoint controller (414) with feedforward and optional feedback (420); runtime witnesses (416); safe-off (418) and recalibration interface (426).

Deployed operating method (ordinary operator; qualification not repeated)

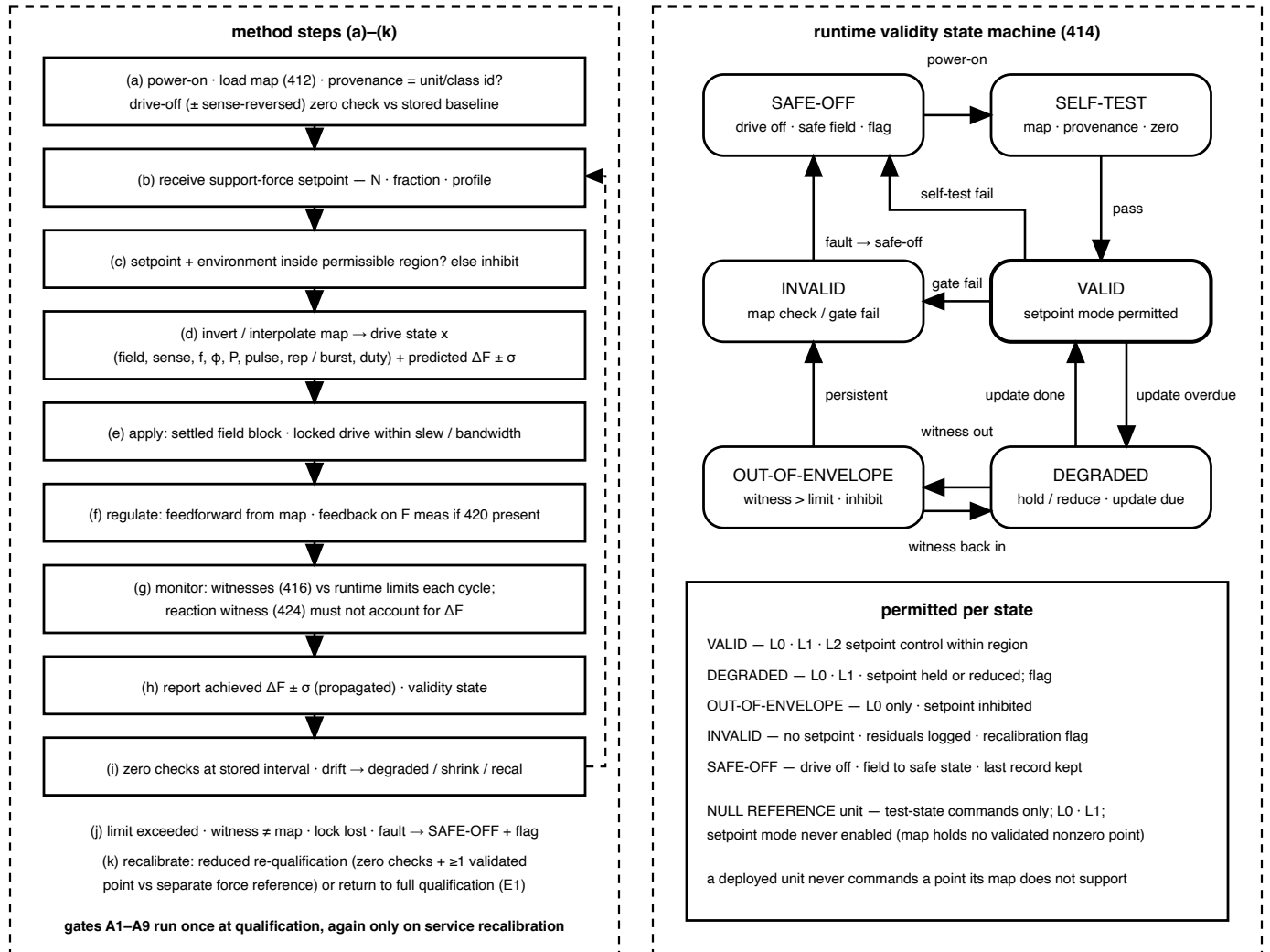


FIG. 2

Deployed operating method — steps (a)–(k) from map load and zero check through setpoint receipt, region check, mapped drive state, regulation, witness monitoring, reporting, interleaved zero checks, safe-off, and recalibration; and the runtime validity state machine (safe-off, self-test, valid, degraded, out-of-envelope, invalid) with the operation permitted in each state.

Qualification → deployment flow — three tiers

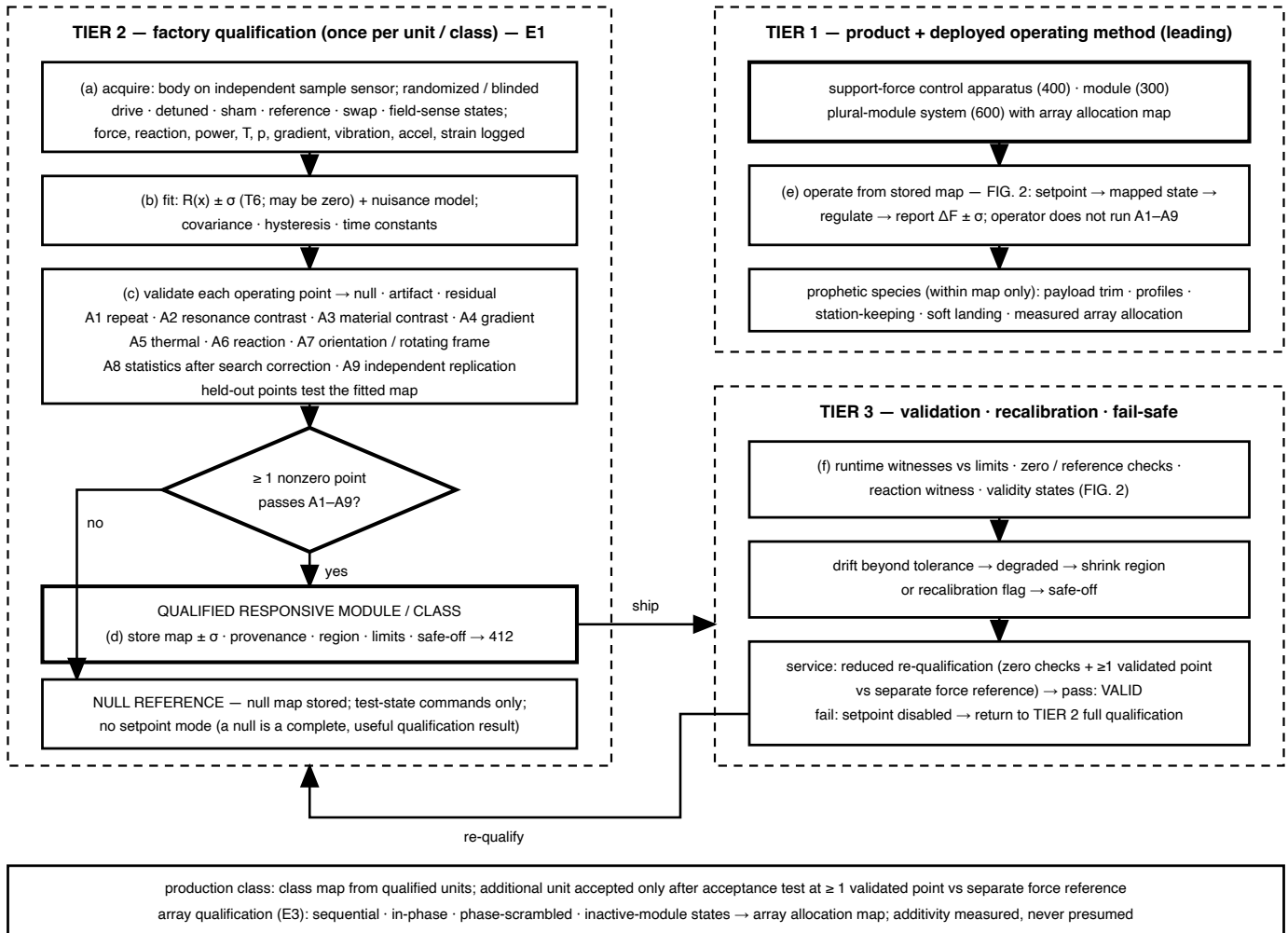


FIG. 3

Qualification-to-deployment flow — Tier 2 factory qualification (acquire, fit, validate under gates A1–A9) classifies a unit as a qualified responsive module or a null reference and stores its map; Tier 1 deployed operation runs from the stored map; Tier 3 runtime validation, recalibration, and fail-safe return a drifting unit to service checks or full re-qualification.

Layered scope — raw observable → validated map → controlled operation

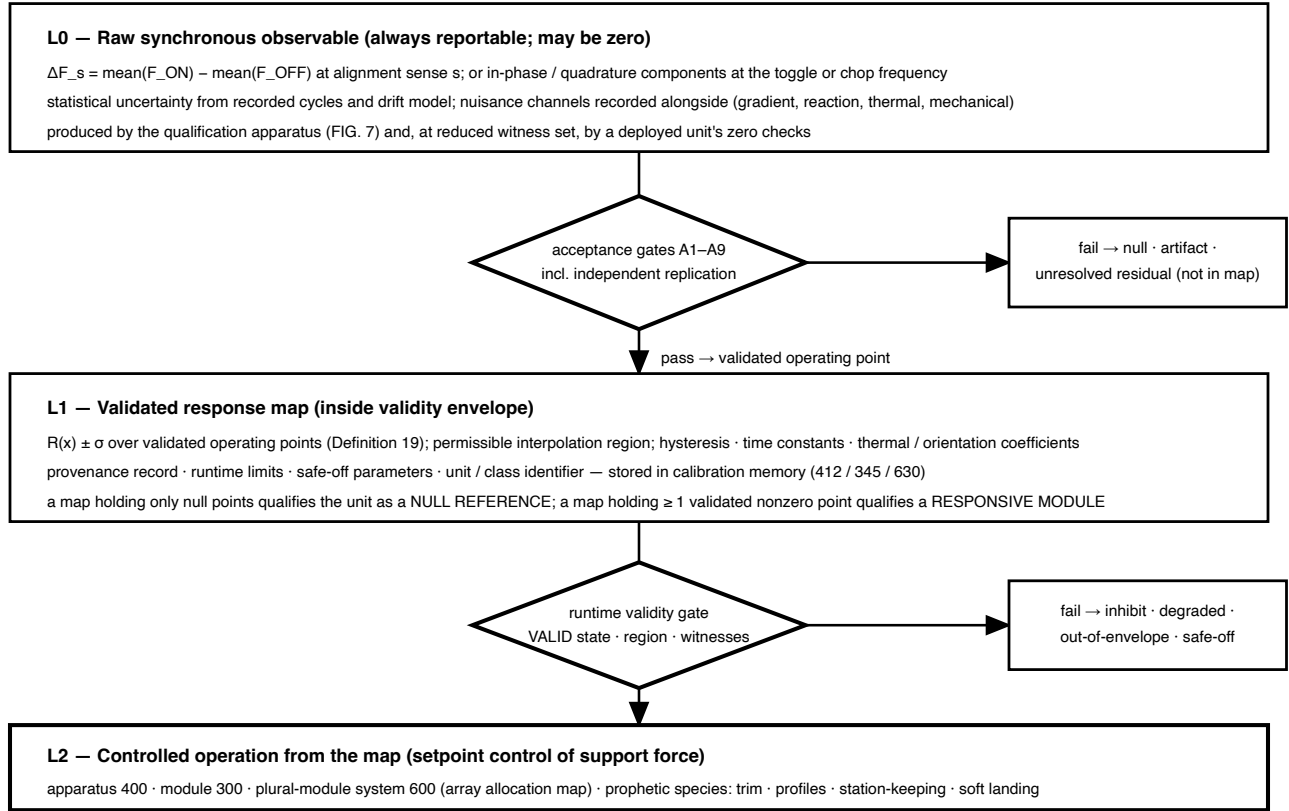


FIG. 4

Layered scope — L0 raw synchronous observable, L1 validated response map, and L2 controlled operation; acceptance gates A1–A9 stand between L0 and L1 (a point enters the map only after passing them) and the runtime validity gate stands between L1 and L2 (a deployed unit commands only validated or permissibly interpolated points in the valid state). No layer presumes sign, magnitude, additivity, or scaling.

Calibrated support-force module (300) — sealed product

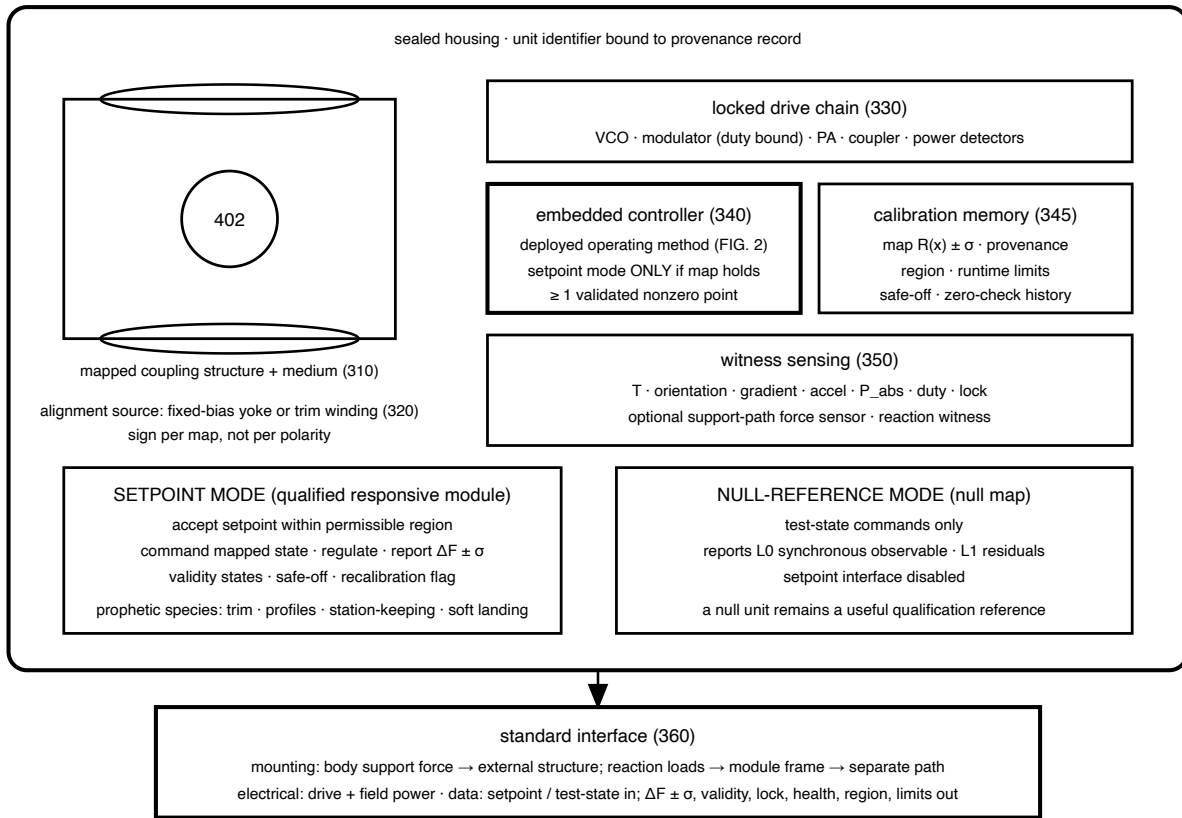


FIG. 5

Calibrated support-force module (300) — sealed unit with mapped coupling structure and medium (310), alignment source (320), locked drive chain (330), embedded controller (340), calibration memory (345), witness sensing (350), and standard mounting/electrical/data interface (360); setpoint mode is enabled only for a qualified responsive module, and a unit holding a null map operates as a null reference.

Plural-module support-force system (600) — per-module maps, measured array allocation

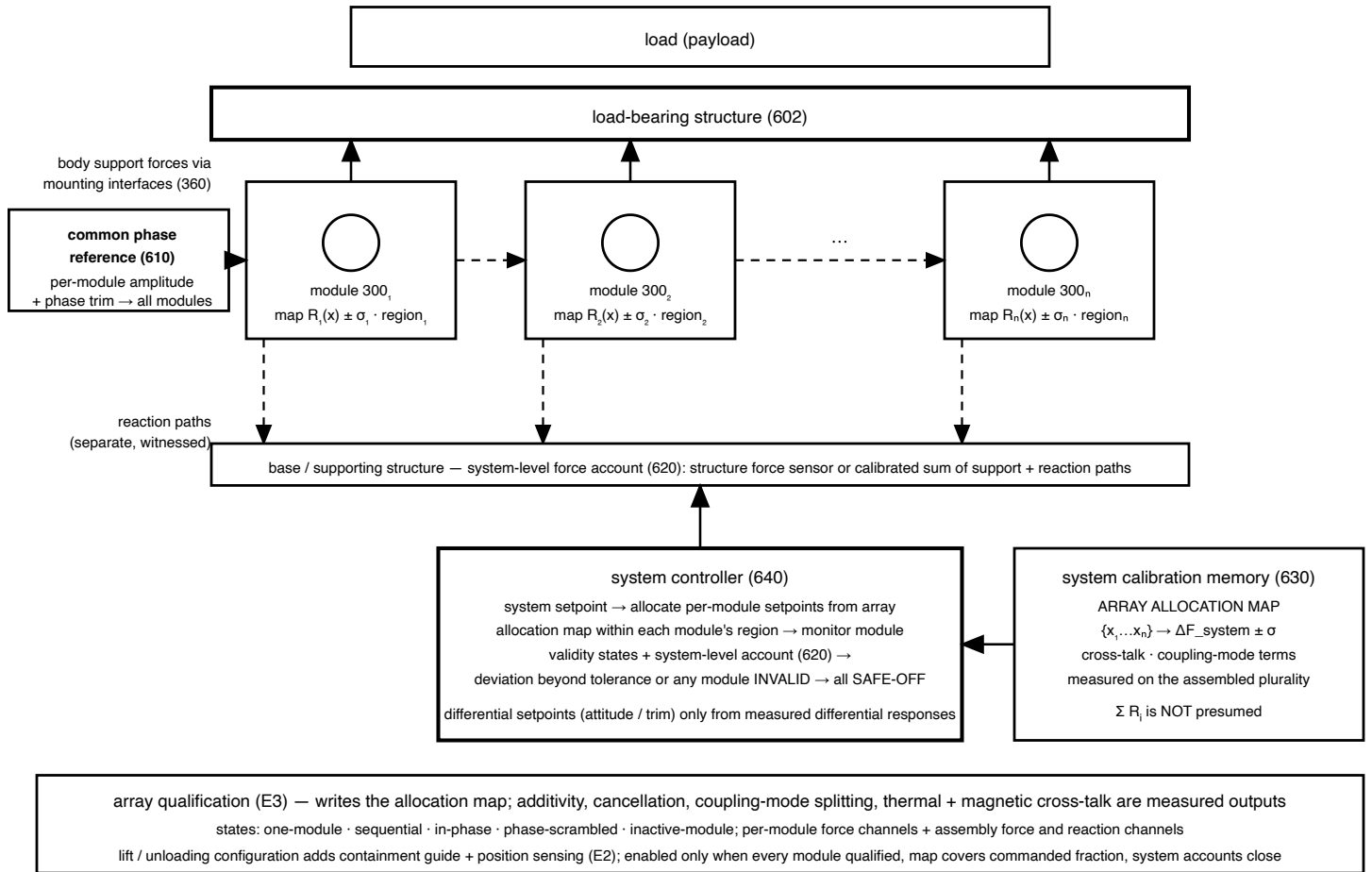


FIG. 6

Plural-module support-force system (600) — modules (300₁ ... 300_n) holding individual response maps bear a common load-bearing structure (602) through their mounting interfaces while reaction paths are separately witnessed; a common phase reference (610), a system-level force account (620), a system calibration memory (630) holding the measured array allocation map, and a system controller (640) that allocates setpoints from that map and drives all modules to safe-off on a system-level deviation or module invalidity; array qualification states listed.

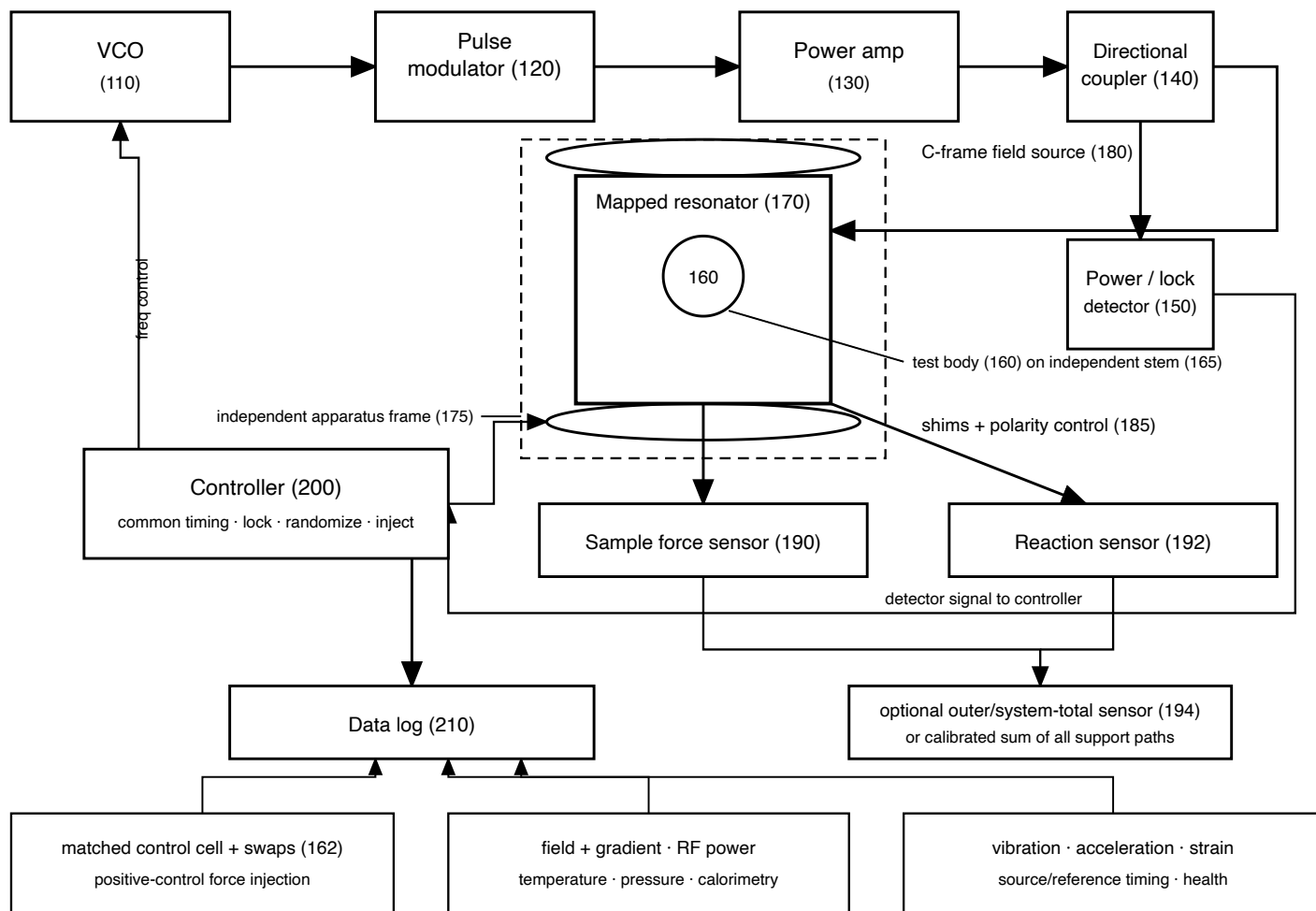
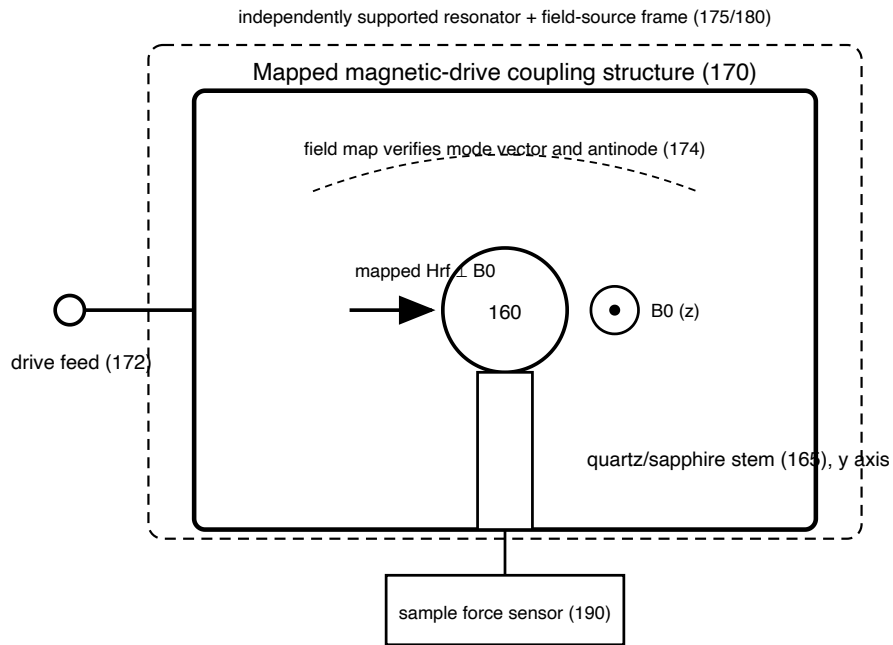


FIG. 7

Qualification apparatus (Tier 2) — null-valid system: sample, source-reaction, and optional system-total force accounts; locked drive; matched control; calibration injection; and synchronized field-gradient, RF-power, thermal, pressure, vibration, acceleration, and strain channels.



reference TE_{102} seed: $a_x \times b_y \times d_z = 47.55 \times 22.15 \times 64.77$ mm at 5.600 GHz

alternatives: TE · loop-gap · re-entrant · waveguide/coil · mapped off-axis mode

the dielectric stem alone loads the sample sensor; resonator and C-frame load the source frame

FIG. 8

Coupling detail — sample at a measured transverse microwave magnetic-field antinode on an independent force path.

nonoverlapping drive pulses; controller enforces registered duty bound



absorption lock (reflected-power minimum)



randomized labeled or blinded state windows

resonant +B	detuned -B	drive OFF	sham +B	resonant -B	phase-scrambled
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force record (zero permitted)



registered contrasts + nuisance model + multiplicity correction; no response shape assumed

FIG. 9
Timing — bounded-duty drive, frequency lock, and randomized differential state windows.

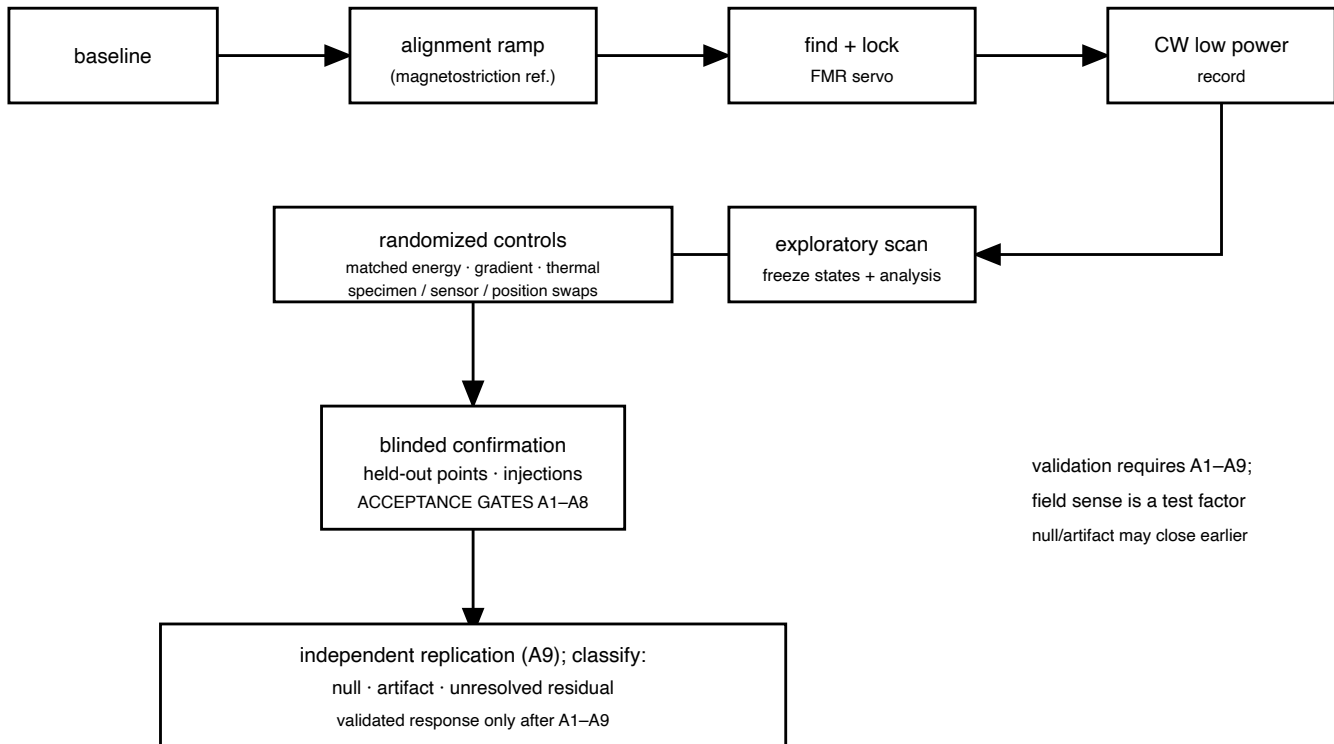


FIG. 10

Qualification controller state machine — exploration, frozen confirmation, matched nuisance controls, acceptance gates A1–A8, and independent replication A9; a null or identified artifact remains a complete classification.

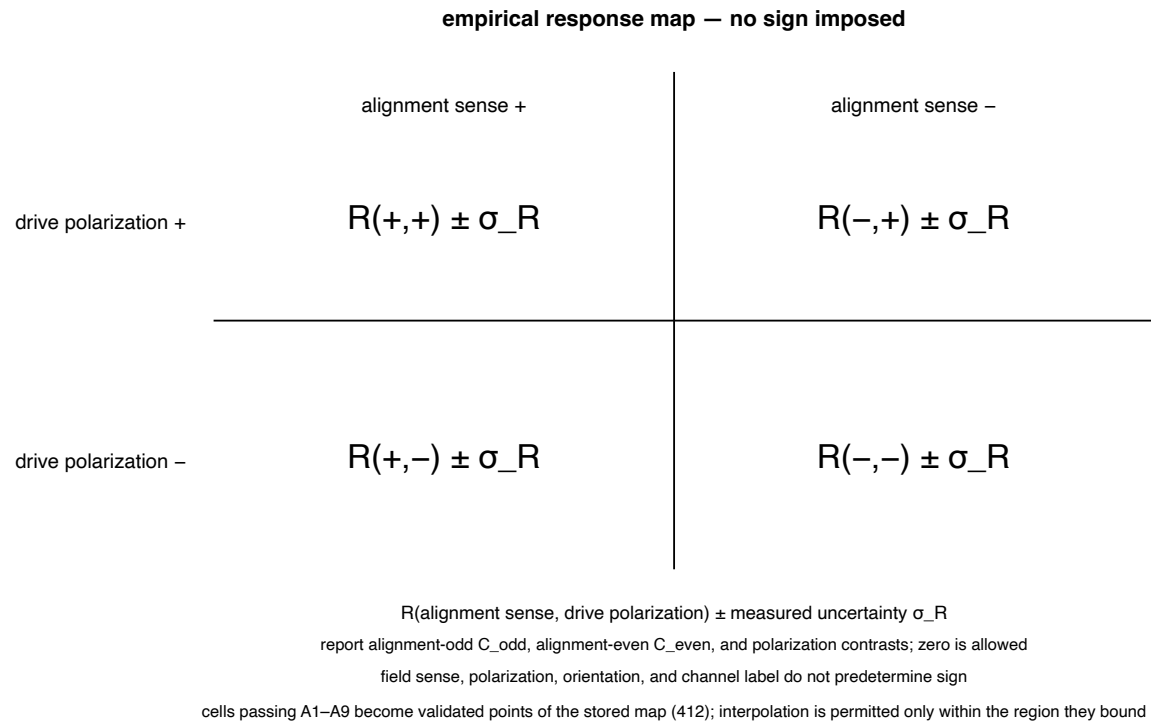


FIG. 11

Empirical response map as stored — the response surface is learned from frozen controls and held-out data rather than imposed by theory; validated cells and the permissible interpolation region they bound are written to the calibration memory as the response map.

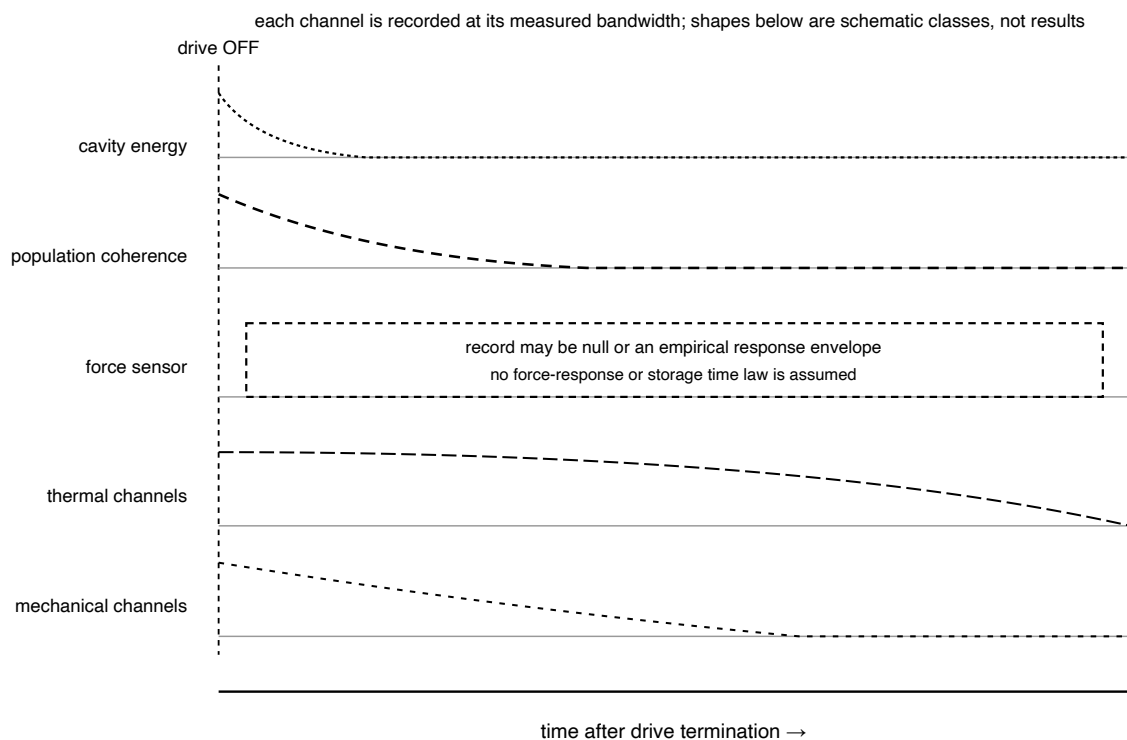


FIG. 12

Multi-timescale diagnostic — cavity, population, force-sensor, thermal, and mechanical channels are recorded at their measured bandwidths; the schematic asserts neither a nonzero force response nor a storage law.

Nested rotating-frame qualification apparatus (700)

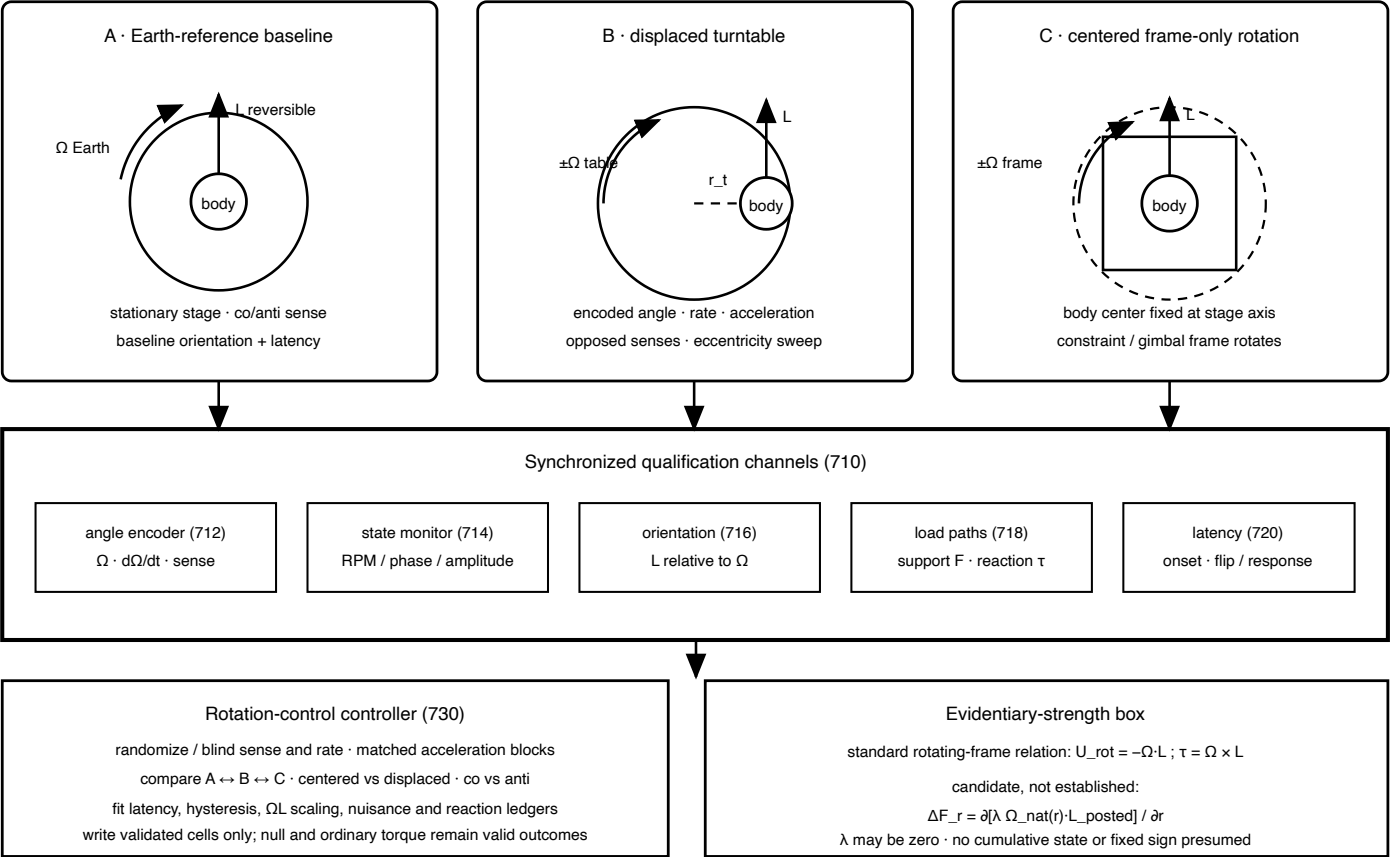


FIG. 13

Nested rotating-frame qualification apparatus (700) — Earth-reference baseline, displaced turntable rotation, and centered constraint-frame rotation; synchronized angular-position/rate/acceleration, state, orientation, support-force, reaction-torque, and latency channels; randomized comparisons under controller (730). The standard rotating-frame energy and torque are distinguished from the candidate, unestablished radial constitutive relation.